ACTUARIES IN SPACE: PRICING THE FRONTIER

2026 SOA Student Research Case Study Challenge

Report to Galaxy General Insurance Company Management

Recommended Products for Cosmic Quarry Mining Corporation

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1. Executive Summary

Galaxy General Insurance Company (herein GGIC) recommends a portfolio of four insurance products in response to Cosmic Quarry Mining Corporation's RFP, providing comprehensive coverage across equipment failure, cargo loss, worker's compensation, and business interruption for operations in the Helionis Cluster, Bayesia System, and Oryn Delta solar systems. Each product is commercially viable and designed to be scalable as Cosmic Quarry's operations grow. All four products are underpinned by GLM-based pricing models fitted to GGIC's historical claims experience, enabling granular, covariate-driven premiums that are portable across solar systems.

1.1 Product Recommendations

- **Cargo Loss:** Single-mission all-risk cover on physical cargo loss/damage. Pricing driven by route risk, container type, debris density, pilot experience, and solar radiation. Gross premium rates range 1.8–4.2% of cargo value, depending on route risk.
- **Equipment Failure:** Cover for repair or replacement of mining equipment from mechanical, system, or environmental failures. Premiums calculated using numerical and categorical factors, but final prices are grouped by solar system and equipment types.
- **Workers' Compensation:** Cover for all work-related injuries and illnesses, with benefits including periodic income payments, medical expenses, and lump-sum permanent impairment payments. This product carries very low aggregate risk, with an expected annual loss per policy of approximately $68 and a 95% VaR of effectively $0. Premiums are driven primarily by occupation and psychological stress level, ranging from $36 for a low-stress administrator to $564 for a high-stress drill operator.
- **Business Interruption:** Annual business interruption coverage with optional catastrophe extension for interstellar mining stations, protecting against loss of operating income and continuing operating expenses following operational shutdowns caused by equipment failure, interruptions, or environmental hazards.

1.2 Solar-System Tailoring & Scalability

Each solar system's distinct risk profile is embedded directly in pricing. Helionis Cluster faces elevated debris risk from outer asteroid clusters despite a stable star. Bayesia System's low debris risk is offset by binary-star radiation spikes. Oryn Delta carries the highest combined hazard, with an asymmetric asteroid ring and unpredictable dwarf star flares.

Solar system is a direct rating variable across equipment failure, workers' compensation, and business interruption models. For cargo loss, route-specific covariates naturally capture each system's environment, allowing new solar systems to be onboarded by declaring relevant environmental characteristics without model re-estimation.

1.4 ESG Considerations

ESG criteria are embedded as practical underwriting conditions rather than aspirational targets. Environmentally, waste-management plans and debris-mitigation compliance are required as conditions of cover, with premium discounts of up to 5% available for verified adoption of lower-impact extraction technologies. Socially, minimum safety training and protective gear standards are policy conditions for workers' compensation (as safety training can reduce claim frequency from up to 50%). Governance controls including third-party compliance verification and sanctions screening apply across all solar systems, reducing moral hazard and supporting GGIC's position as a responsible insurer in the interstellar market.

1.5 ERM Integration

The four products function to complement Cosmic Quarry's own operational controls. Cargo loss sits alongside route risk governance; equipment failure rewards preventative maintenance through lower premiums; workers' compensation creates a direct financial incentive for stress reduction and safety investment. Business interruption is the most significant cover, with tail exposure reaching $21M at the 99.5th percentile, it should be paired with documented contingency and recovery protocols within Cosmic Quarry's ERM framework. Across all four products, the pricing structure creates direct feedback between operational behaviour and insurance cost, supporting ERM discipline rather than creating moral hazard.

2. Product Design

2.1 Cargo Loss

Coverage: Single-mission, all-risk cover on physical loss or damage of declared cargo. Attaches at cargo handover to the mission operator; ceases at confirmation of delivery. Total loss triggered when recovery is infeasible or recovery cost exceeds 80% of the agreed cargo value.

Triggers: Telemetry-confirmed physical loss/damage (where available); missing or damaged cargo declared upon arrival or identified during arrival inspection.

Deductibles: Cause-based 3-tiered structure: Total vehicle loss/catastrophic failure > Vehicle survived, cargo lost > Vehicle survived, cargo damaged. The deductible amount represents expected administration and investigation costs, with vehicle loss (requiring space investigation) having the highest deductible.

Exclusions: Undeclared route deviation; false/non-disclosed route characteristics, solar events, or debris fields; inherent vice or cargo not certified for container; war, confiscation, politically motivated interference; willful misconduct.

Other features: All rating factors (cargo value, route risk, container type, debris density, pilot experience, solar radiation) must be declared pre-launch. No retrospective declarations. Cargo must be in a certified container type.

2.2 Equipment Failure

Coverage: Multi-mission cover on mechanical or electrical failure of insured mining equipment used in operations. All equipment is insured. Coverage applies to repair or replacement costs of damaged machinery and associated diagnostic or recovery costs. Coverage attaches when equipment is commissioned for operation and ceases once the asset is repaired, replaced, or decommissioned.

Triggers: Telemetry-confirmed mechanical, electrical, or control-system failure that causes the equipment to cease functioning or operate outside predefined and agreed-upon safe parameters. Triggers instantly upon failure of equipment.

Deductibles: Cause-based 3-tiered structure varying by equipment type and usage intensity: high-risk conditions (heavy use/harsh environments) $1k–$2k > moderate conditions $500–$1k > low-risk, well-maintained equipment up to $500.

Exclusions: Wear and tear, inadequate maintenance, undeclared modifications, manufacturer defects, and wilful misconduct.

Other features: All key risk characteristics (equipment type, operational intensity, maintenance schedule, and solar system environment) must be declared before operation.

2.3 Workers' Compensation

Coverage: All injuries at work or sicknesses developed due to direct workplace exposure, including injuries from operating equipment, space radiation, motion sickness or physical strain from low gravity levels.

Triggers: When a worker experiences an injury or illness. This is to be supported with medical documentation and reports to confirm they occurred during work.

Benefit structure: Periodic payment to cover lost earnings, medical expenses, property damage payments, work injury damages, and lump sum payments for permanent impairment.

Exclusions: Injuries occurred outside of work, self-harm, or illegal activities.

2.4 Business Interruption

Coverage: The policy provides coverage for loss of operating income and continuing operating expenses, including wages, equipment leases, and maintenance contracts, following an insured business interruption event. Covers all energy backup levels for all solar systems and stations.

Triggers: Coverage triggered by operational disruptions such as mechanical or equipment failures, energy system failures, supply chain interruptions, environmental hazards (e.g., solar storms), or infrastructure failures that result in shutdown of mining operations. Must have waiting periods of 24-72 hours of business interruption for business interruption to be claimable.

Deductibles: Cause-based 3-tiered structure: Weak systems / harsh environments, with $250k-$500k deductible > moderate conditions, with $100k-$250k deductible > high resilience systems with strong backup systems, with $50k-$100k deductible. Reasoning for the deductible is that the expected loss is $443k / year, thus it is severe but infrequent

Exclusions: The policy excludes losses arising from intentional operational shutdowns, planned maintenance, negligence or failure to follow safety protocols, and catastrophic events deemed uninsurable.

Other features: Periodic policy reviews and environmental compliance checks are conducted annually or biannually to ensure adherence to operational protocols.

3. Summary of Pricing

Note that the premiums used in this analysis are not directly comparable for long-term estimation purposes due to differences in their underlying exposure periods. Further, the cargo loss premium is charged per trip, whereas the other three coverages are priced annually. Consequently, these premiums are only suitable for the short term and require adjustment for exposure frequency for potential long-term purposes. Please refer to the appendix for further details regarding model design.

3.1 Cargo Loss

Frequency Model: Negative Binomial GLM

Claim counts per shipment are overdispersed (variance/mean ratio of ~4.5), thus unsuitable for a Poisson model. This frequency was subsequently modelled using a Negative Binomial GLM, with parameter selection performed by minimising AIC and retaining statistically significant covariates. Distance, transit duration, vessel age, weight, and cargo type were non-significant.

Specific route hazards (debris density, solar radiation, route risk) are significant predictors of claim rate in addition to crew experience and container type, while distance and transit duration were non-significant. This suggests that individual route hazards (takeoff, landing, asteroid belts) are the key cause of claims, and time spent travelling through open void does not pose a significant risk.

Severity Model: Beta Regression

Severity ratio (claim amount ÷ cargo value) modelled with Beta regression, enforcing the restriction that claim amount ≤ cargo value. Similarly, parameter selection involves minimising AIC while removing statistically insignificant covariates. Key factors for severity modelling were levels of route risk, debris density, and solar radiation. Notably, crew experience and container type do not contribute to claim severity, suggesting that the severity and nature of accidents are dependent only on route-specific factors.

Indicative Rating Table

Gross rates below apply loadings of 15% expense + 10% profit + 5% risk (1.43× multiplier) at mean covariate profiles per route risk:

Route Risk	E[Freq]	E[Sev]	Pure Rate	Gross Rate	Prem on $10M insured
1	0.203	6.3%	1.28%	1.82%	$182K
2	0.228	7.5%	1.71%	2.43%	$243K
3	0.242	8.7%	2.11%	3.02%	$302K
4	0.257	9.7%	2.49%	3.57%	$357K
5	0.263	11.1%	2.92%	4.17%	$417K

3.2 Equipment Failure

Frequency Model: Poisson GLM

The frequency of equipment failure claims was modelled using a Poisson GLM. The dependent variable was the number of claims observed in each exposure period. Log-exposure was included as an offset as a linear relationship was identified between operational time and claim frequency. After continuous predictor standardisation to improve interpretability and stability, key significant predictors were equipment age, maintenance intensity, usage intensity, equipment type, and solar system type.

Model diagnostics suggest a good Poisson fit with deviance dispersion of 0.38 and Pearson dispersion of 1.10. Further, this parsimonious model achieved a relatively low AIC compared to simpler models tested.

Severity Model: Gamma GLM

Claim severity was modelled using a Gamma GLM, appropriate for the positive, right-skewed distribution of repair costs. The model achieved a strong fit ($R^2 = 0.296$), with equipment type and solar system location identified as the primary drivers of claim costs. Older equipment and higher usage intensity were also associated with increased severity, reflecting accumulated wear and operational strain. Full model diagnostics are provided in Appendix 1.

Indicative Rating Table

Technical premiums are found at loading levels of 15% expense + 10% profit.

Solar System	Equipment Type	Expected Claim Frequency	Expected Claim Severity ($)	Pure Premium ($)	Technical Premium ($)
Epsilon	Flux Rider	0.20	60,003.45	12,794.24	15,992.80
Epsilon	Graviton Extractor	0.16	74,244.57	12,506.67	15,633.34
Epsilon	Ion Pulverizer	0.15	84,180.62	13,252.28	16,565.35
Epsilon	Quantum Bore	0.13	119,305.24	16,601.95	20,752.43
Epsilon	Reglaggregators	0.13	82,009.90	11,578.16	14,472.69
Helionis Cluster	Flux Rider	0.30	54,473.32	17,058.96	21,323.70

Helionis Cluster	Graviton Extractor	0.23	67,400.41	16,520.43	20,650.54
Helionis Cluster	Ion Pulverizer	0.22	80,996.19	17,819.16	22,273.95
Helionis Cluster	Quantum Bore	0.19	115,353.45	21,917.16	27,396.45
Helionis Cluster	Reglaggregators	0.20	76,666.16	15,333.23	19,166.54
Zeta	Flux Rider	0.19	69,487.04	13,202.54	16,503.17
Zeta	Graviton Extractor	0.15	83,273.28	12,844.26	16,055.33
Zeta	Ion Pulverizer	0.14	94,426.64	13,642.15	17,052.68
Zeta	Quantum Bore	0.12	133,976.54	17,078.83	21,348.54
Zeta	Reglaggregators	0.12	92,085.42	11,987.49	14,984.36

3.3 Workers' Compensation

Frequency Model: Poisson GLM

Claim frequency was modelled using a Poisson GLM, selected over a Negative Binomial model based on a lower AIC. The model achieved a strong fit with a Pearson dispersion of 0.99. Occupation, accident history, psychological stress level, and safety training index were identified as significant predictors of claim frequency, confirming that workplace risk characteristics are the primary drivers. Solar system, employment type, and experience years were not significant. Full model details are provided in the appendix.

Severity Model: Log-Normal OLS Regression

Claim severity was modelled using a Log-Normal OLS regression, selected based on lowest AIC across distributions tested. Occupation, psychological stress level, gravity level, safety training index, and protective gear quality were significant predictors. Higher stress and more hazardous roles were associated with larger claims, while stronger safety measures were associated with lower severity.

Premium Summary

Technical premiums are loaded at 15% expense and 10% profit. Premiums increase with both occupation risk and psychological stress level. The median technical premium is $151.77, rising to $563.79 for drill operators at maximum stress and falling to $35.86 for administrators at minimum stress. This range reflects the meaningful difference in workplace risk across Cosmic Quarry's workforce.

3.4 Business Interruption

Claim frequency and severity were modelled separately, with expected losses calculated as their product and premiums derived by applying expense, profit, and risk loadings.

Route risk was segmented by solar system and energy backup resilience into low (1-2), medium (3), and high (4-5) tiers, reflecting infrastructure resilience as the primary driver of business interruption risk.

Frequency Model: Negative Binomial GLM

A Negative Binomial GLM was selected due to overdispersion in claim counts (dispersion ratio 1.73), where a Poisson model would underestimate variability. Solar system, energy backup score, and key operational characteristics were retained as significant predictors following variable selection at a 5% significance threshold. Full model diagnostics are provided in the appendix.

Severity Model: Gamma GLM

Claim severity was modelled using a Gamma GLM with log-link, appropriate for the strictly positive, right-skewed nature of business interruption losses. The final model retained the solar system, energy backup score, and safety compliance as significant predictors. The pseudo R^2 of 0.00088 indicates limited explanatory power over severity, reflecting the inherently unpredictable nature of large interruption losses. Full diagnostics are provided in the appendix.

Indicative Rating Table

Gross rates below apply loadings of 15% expense + 10% profit + 5% risk (1.43× multiplier) at median covariate profiles per energy backup level:

Solar System	Backup Risk Tier	Expected Claim Frequency	Expected Claim Severity ($)	Pure Premium ($)	Technical Premium ($)
Epsilon	Moderate	0.195	4,003,731	780,142	975,177
Epsilon	Strong	0.215	4,440,526	953,601	1,192,001
Epsilon	Weak	0.201	4,563,151	918,454	1,148,068

Helionis Cluster	Moderate	0.208	5,200,253	1,080,615	1,350,769
Helionis Cluster	Strong	0.197	4,556,111	896,505	1,120,631
Helionis Cluster	Weak	0.183	4,659,942	852,032	1,065,040
Zeta	Moderate	0.199	3,577,193	710,934	888,668
Zeta	Strong	0.216	4,424,077	954,511	1,193,139
Zeta	Weak	0.188	4,095,042	770,383	962,979

4. Capital Modelling

To assess the financial risk associated with each coverage, stochastic aggregated loss models were adopted for equipment failure, cargo loss, business interruption and workers' compensation. A compound frequency-severity framework was used to model the total annual loss: $S = \Sigma_{i=1}^{N} X_i$, where N represents the frequency, that is, the number of claims occurring in a year, and X_i represents the individual claim severity.

Historical claim data were used to estimate the parameters of frequency and severity distributions for each line of business. Monte Carlo simulation with 100,000 iterations was performed to generate aggregate annual loss distribution, and estimate key risk metrics, including expected loss, variance and Value-at-risk for tail risk. The frequency and severity models used are summarised below.

Line of Business	Frequency Model	Frequency Parameters	Severity Model	Severity Parameters
Equipment Failure	Poisson	λ = 0.1839667	Gamma	shape = 4.32, scale = 11,100
Business Interruption	Negative Binomial	r = 0.121553, p = 0.547205	Gamma	shape = 0.3538, scale = 12,327,906
Cargo Loss	Negative Binomial	r = 0.5370, p = 0.6868	Beta × cargo value	α = 1.3832, β = 14.3850
Workers' Compensation	Poisson	λ = 0.017665	Lognormal	μ = 7.619821, σ = 1.0786

4.1 Aggregate Loss Distributions for Each Hazard Coverage

The simulated loss distributions for all four products are highly right-skewed, aligning with the nature of large, aggregate insurance losses with low frequency but high severity.

Metric	Equipment Failure	Cargo Loss	Business Interruption	Workers' Compensation
Expected annual loss ($)	8,935	1,978,730	443,322	67.93
Variance	529,525,300	129,887,900,000,000	8,752,651,271,890	890,314
Standard deviation ($)	23,011	11,396,840	2,958,488	943.56
VaR (95%) ($)	61,858	6,597,604	596,933	0
VaR (99%) ($)	103,491	57,528,540	13,684,392	1,672.69
VaR (99.5%) ($)	120,090	83,444,880	21,261,136	3,720.63

Cargo Loss and Business Interruption stand out due to their high tail risk, accompanied by their significant expected losses of $1.98 million and $443k million per annum with high volatility. At 99.5% VaR, Cargo Loss exhibits the largest loss of $83 million, followed by Business Interruption's $21 million, compared to Equipment Failure and Workers' Compensation's relatively negligible tail risk.

The disparity reflects the underlying exposure and insured value as a driver of loss. The cargo loss is attributed to the high value of the shipment during transit that poses huge potential losses under extreme scenarios. Business Interruption exhibits moderate tail risk from the consequences of large-scale operational shutdowns. In contrast, equipment failure and workers' compensation contribute minimal losses due to their predictable claim patterns, and lower variability and claim severity, leading to less extreme outcomes to the overall portfolio exposure.

Net Revenue and Returns

At expected loss conditions, net revenue is estimated as the 10% profit loading on the pure premium. These are short-term baseline figures, subject to the 2.46% annual inflation adjustment in section 4.3 for long-term projections.

Here, net revenue is calculated assuming similar policy exposure to historical GGIC experience, totalling over $25 billion annually.

Product	Policy Count	Expected Annual Loss ($m)	Net Revenue ($m)
Cargo Loss	120,000	237,447.6	23,744.76
Business Interruption	100,000	44332.2	4,433.22
Equipment Failure	90,000	804.15	80.42
Workers' Compensation	130,000	8.83	0.88

4.2 Stress testing for extreme scenarios

Stress testing was performed to assess the sensitivity to loss of each product in adverse scenarios. In addition to the baseline scenario, which is based on the empirical losses observed and analysed in the previous section, moderate and extreme stress scenarios of assumptions were also applied to test the tail behaviour.

Scenario	Assumptions
Baseline	Fitted model assumptions and parameters
Moderate Stress	Claim frequency increased by 25%, Claim severity increased by 10%
Extreme Stress	Claim frequency increased by 50%, Claim severity increased by 30%

Expected Annual Loss ($)

Product	Baseline	Moderate Stress	Extreme Stress
Equipment Failure	8,803	12,788	17,207
Business Interruption	2,037,357	2,781,299	3,763,331
Cargo Loss	443,322	648,226	945,169
Workers' Compensation	68	92	118

Value-at-Risk (99%)

Product	Baseline	Moderate Stress	Extreme Stress
Equipment Failure	102,773	129,148	157,252
Business Interruption	59,961,210	76,315,150	94,751,300
Cargo Loss	13,684,390	18,918,430	26,962,710
Workers' Compensation	1,673	2,494	3,430

Under moderate and extreme stress conditions, expected losses increase significantly across all products. However, the dollar amount impact is the greatest for Cargo Loss and Business Interruption. The 99% VaR for business interruption increased from approximately $60 million to $95 million, and Cargo Loss increased from $13.7 million to $27 million. The results demonstrate the high sensitivity of tail-risk, especially for products with high claim severity like Business Interruption and Cargo Loss, as small unfavourable changes in assumptions will disproportionately increase the extreme losses.

Though all products are sensitive to unfavourable changes in assumptions, the results signal that a larger capital amount is required to capture the extreme potential losses driven by Business Interruption and Cargo Loss under stress, to comply with the regulatory requirement and internal risk tolerance.

4.3 Inflationary Pressure

Inflation introduces systematic upward pressure on claim severity and exposure. Inflation was incorporated directly within the Monte Carlo simulations through severity and exposure adjustments. Simulated claim severities were multiplied by an inflation factor of 2.46% (15-year average inflation rate) to reflect expected increases in repair costs, replacement costs and insured asset values. This effectively introduced a forward-looking cost basis. For products such as Cargo Loss and Business Interruption, where losses are proportional to underlying asset values, this also captures the implicit growth in the sum insured. The stress testing scenarios were also calibrated to include inflationary effects by applying higher severity factors under adverse conditions, since inflation may accelerate during periods of economic stress. This would help ensure that the mean loss and tail risk VaR are adjusted for inflation.

5. Risk Assessment

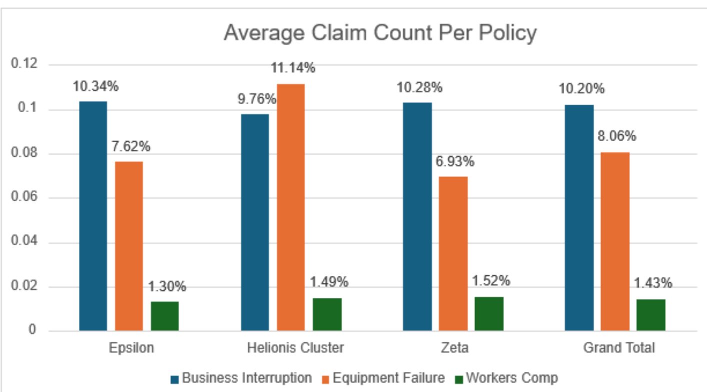
5.1 Risk identification by hazard area coverage for each solar system

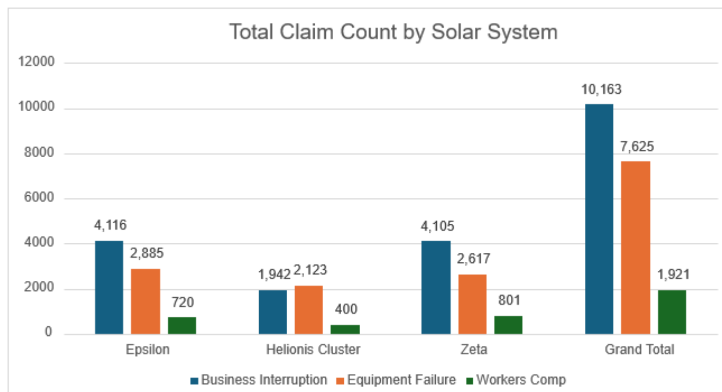
The three solar systems present distinct risk profiles across the four hazard areas, driven by their stellar and orbital environments.

Helionis Cluster benefits from a stable star with low radiation exposure, moderating workers' compensation severity and radiation-linked equipment degradation. However, two outer-system asteroid clusters elevate debris density, making cargo loss and equipment failure the primary concerns. This is reflected in the pricing model, where Helionis Cluster carries the highest equipment failure claim frequencies across all equipment types. Business interruption exposure is moderate but costly when triggered, with an average claim size of $4,837,695.

Bayesia System presents the inverse pattern. Well-mapped asteroid belts keep cargo loss and debris-related equipment failure risk lower, but binary stars produce sharp radiation spikes that elevate workers' compensation frequency and equipment failure severity. Business interruption risk is meaningful given that radiation events can trigger simultaneous equipment shutdowns across multiple stations.

Oryn Delta carries the highest combined risk profile. An asymmetric asteroid ring drives persistent debris exposure across cargo and equipment lines, while a dim dwarf star introduces unpredictable flares that compound radiation risk. Unlike Bayesia's patterned radiation events, Oryn Delta's sporadic flare activity makes loss forecasting more uncertain, which is particularly consequential for business interruption where operational disruptions are harder to anticipate.





From data visualisations of historical experience, business interruption has a grand total of 10,163 claims, more than 7,625 of which are equipment failure, and 1921 which are workers' compensation. From looking at the average claim count per policy, business interruption also has the highest probability of claim, with 10.2%, whereas equipment failure has 8.06% and workers' compensation has 1.43%. Business interruption also has the largest number of claim entries, with 10044, with equipment failure having 8257, followed by workers' compensation having only 1915. Additionally, business interruption has the largest average claim size across all solar systems.

From these statistics, we can say that across all three solar systems, business interruption is the most hazardous, as when it occurs, it is the most severe, and has the largest number of claims in Epsilon and Zeta. It also has the highest average claim count per policy in these two solar systems. Equipment failure, however, has the highest average claim count per policy, with 11.14% compared to 9.76% for business interruption. Despite this, the average claim size of the Helionis cluster for business interruption far exceeds equipment failure, with a $4,837,695 claim size vs an average of $78,863 for equipment failure. Therefore, business interruption is the most severe across all solar systems, followed by equipment failure, then workers' compensation.

5.2 Correlated Risk Scenarios

In modelling, there was the assumption of independence between solar systems, however, this assumption may not hold under systemic events. Common external shocks such as solar storms introduce positive dependence between claim frequency and severity, as multiple solar systems may simultaneously experience disruptions. Aggregate losses may therefore be underestimated if correlation is overlooked.

The three solar systems differ meaningfully in their vulnerability to correlated events. Bayesia System faces the greatest exposure to radiation-driven correlation, as its binary stars can produce simultaneous radiation spikes that elevate equipment failure and workers' compensation claims across all stations concurrently. Oryn Delta carries compounding risk, where dwarf star flares combined with asteroid disturbances could simultaneously trigger cargo loss, equipment failure, and business interruption across multiple sites. Helionis Cluster, while more stable, remains exposed to correlated debris events given its two outer asteroid clusters, which could affect multiple transit routes simultaneously.

Although the magnitude of impact would differ across systems, a galactic solar storm or large-scale debris event would affect all three regions concurrently, and losses would be positively correlated. This reduces diversification benefits and leads to a heavier-tailed aggregate loss distribution than the independence assumption implies. Practically, this means the 99.5% VaR figures reported in section 4 should be treated as lower bounds under extreme correlated scenarios, with business interruption and cargo loss most exposed given their high individual tail risk.

5.3 Cargo Loss — Key Risks

Threats	Likelihood	Severity	Rating	Mitigation
Solar storm/flare	Moderate	Very High	Critical	Exclusion clause: Solar exclusion is predicted but undisclosed
Debris field collision	Mod–High	High	High	Route declaration; density rating
Total vessel loss	Low	Catastrophic	High	Tier 1 deductible; Stop-loss reinsurance (1)
Container failure	Moderate	Moderate	Medium	Container certification
Pilot error	Moderate	Moderate	Medium	Experience rating factor
Route deviation	Low	Variable	Medium	Exclusion clause
Willful misconduct	Low	Variable	Low–Med	Exclusion clause; identified via investigation

Risk varies by solar system through environmental covariates: systems with denser asteroid belts have higher debris density on transit routes; systems closer to active stars face elevated solar radiation; route risk classifications reflect localised navigational hazards.

(1) Stop-loss reinsurance was selected due to the high-tailed risk observed in capital modelling, over alternatives like proportional reinsurance

5.4 Equipment Failure — Key Risks

Threat	Likelihood	Severity	Rating	Mitigation
Mechanical component failure	Moderate	High	High	Preventative maintenance requirements
Control system / AI malfunction	Low–Moderate	Very High	High	Diagnostic monitoring and system redundancy
Radiation damage to electronics	Moderate	High	High	Radiation shielding and solar system risk rating
Structural fatigue from heavy extraction	Moderate	High	High	Equipment age rating and inspection requirements
Debris impact on equipment	Low–Moderate	High	Medium–High	Protective shielding and environmental risk pricing
Poor maintenance practices	Moderate	Moderate	Medium	Maintenance documentation and compliance requirements

5.5 Workers' Compensation — Key Risks

Threat	Likelihood	Severity	Rating	Mitigation
On-site injury (slips, trips, falls)	Moderate	High	High	Safety training, protective gear, and workplace inspections
Exposure to hazardous materials	Low–Moderate	High	High	Personal protective equipment and monitoring protocols
Fatigue-related accidents	Moderate	Moderate	Medium	Shift scheduling, rest periods, and workload monitoring

Gravity or environmental hazards	Low–Moderate	High	Medium–High	Environmental controls, gravity-adapted equipment, and hazard awareness training
Equipment misuse by staff	Moderate	Moderate	Medium	Standard operating procedures and supervision
Wilful misconduct	Low	Variable	Low–Medium	Policy exclusions and investigation procedures

5.6 Business Interruption — Key Risks

Threat	Likelihood	Severity	Rating	Mitigation
Major equipment malfunction	Medium	High	High	Preventative maintenance programs, equipment redundancy, and predictive monitoring systems
Solar storm / extreme environmental event	Low–Medium	Very High	High	Hardened infrastructure, radiation shielding, and early warning space weather systems
Energy system failure	Medium	High	Medium-High	Backup power systems, redundant energy grids, and energy storage capacity
Supply chain disruption	Medium	Medium–High	Medium	Diversified suppliers, inventory buffers, and alternative transport routes
Infrastructure failure (station systems)	Low–Medium	High	Medium	Infrastructure inspections, redundancy in critical systems, and scheduled upgrades
Operational shutdown due to a safety incident	Low	Medium	Low-Medium	Strict safety protocols, worker training programs, and automated monitoring systems

5.7 Scenario Testing (All Hazards)

Scenario	Description	Cross-Hazard Impact
Best	Smooth operations; attritional claims; median covariates	Losses near expected across all lines; comfortable margins
Moderate	Elevated debris/radiation on key routes; some large claims	Above-expected losses in cargo + equipment; within capital
Worst	Galactic solar storm; multiple vessel losses; 99th percentile conditions	Tail losses across all four lines simultaneously; reinsurance triggered

6. Assumptions

6.0 Global Assumptions

1. **No correlation between solar systems for models**: Claims are assumed independent across solar systems. This is a simplifying assumption acknowledged to be a lower bound on tail risk under correlated shock scenarios such as galactic solar storms.

6.1 Cargo Loss

1. **Claim independence:** Claims within shipments are independent; NB model accounts for cross-shipment heterogeneity.
2. **Severity < cargo value:** Claim amount strictly less than cargo value (severity ratio $\in (0,1)$), justifying Beta regression. Any replacement or consequential costs beyond this are explicitly excluded, reducing moral hazard.
3. **Stationarity:** Historical covariate–loss relationships assumed to be stable over the projection period. No long-run trend data exists to challenge this.
4. **Loadings:** Expense 15%, profit 10%, risk 5% applied multiplicatively to pure premium to reflect high tail risk.
5. **Non-significant covariates excluded:** Cargo type, distance, transit duration, vessel age, and weight dropped after testing. Risk is driven by route hazard characteristics, not route length.
6. **Container grouping:** DeepSpace Haulbox + DockArc Freight Case merged into reference level (non-significant individually).

6.2 Equipment Failure

1. **Claim Capping:** All claim amounts above $790k capped at $790K (upper bound specified in the data dictionary).

2. **Loadings:** Expense 15%, profit 10%. No risk loading as expected losses are relatively low compared to other products.
3. **Significance Level:** 5% level used to accept or reject covariates
4. **Equipment Type:** Use the equipment types in the data (slight discrepancy between the data dictionary and the actual data)

6.3 Workers' Compensation

1. **No Policy Limit:** Claims severity is assumed not to be affected by policy limits or deductibles.
2. **Loadings:** Expense 15%, profit 10%. No risk loading as expected losses are very low, 95% VaR over the next year is $0.
3. **Significance Level:** 5% level used to accept or reject covariates
4. **Non-significant covariates excluded:** solar system, employment type, experience years, hours per week, supervision level, base salary, injury type, and injury causes.

6.4 Business Interruption

1. **Claim independence:** Claim occurrences and claim severities are independent across stations and time periods. NB model accounts for cross-station heterogeneity.
2. **Stationarity:** Historical dataset between operational variables and claims experience will hold in future operations.
3. **Loadings:** Expense 15%, profit 10%, risk 5% applied multiplicatively to pure premium.
4. **Non-Significant covariates excluded:** Variables with p-values less than 5% were removed. Coefficients that were negligible and small enough were similarly removed.
5. **No major systematic shock events in baseline modelling:** The baseline model assumes normal operational conditions and does not explicitly incorporate rare systemic events.

7. Conclusion

Overall, it is recommended that GGIC offer a portfolio of four insurance products across four hazard areas: equipment failure, cargo loss, workers' compensation, and business interruption. This portfolio is financially viable and is expected to generate a net profit of over $25 billion annually. Premiums for cargo loss are charged per trip, whereas the other three coverages are priced annually. Distributions for all four products are highly right-skewed, with cargo loss and business interruptions exhibiting high tail risk. As such, GGIC must maintain adequate capital against the high risks of both these products in line with regulation. Risk exposure and capital requirements can be minimised through the use of reinsurance as discussed above. Appropriate risk mitigation strategies such as safety protocols or inspections are also recommended.

8. Data and Data Limitations

8.1 Data Sources

- **Historical claims:** Four datasets from GGIC's analogous lines (equipment failure, cargo loss, workers' compensation, business interruption).
- **New business data:** Cosmic Quarry's prospective exposure, environmental conditions, and financial assumptions.
- **Supplementary:** Data Dictionary, Online Encyclopedia entries (solar systems, company profiles).

8.2 Key Limitations

- **Cargo Loss:** ~6% of records removed due to missing values and corrupted categoricals. Historical data lacks solar-system identifiers. Transferability from analogous lines to Cosmic Quarry's specific operations is uncertain.
- **Equipment Failure**: The dataset required cleaning steps, including removal of invalid observations and capping of extreme values, resulting in a loss of ~1% of the provided data. Further, the limited available risk variables may introduce bias. The correlations between risks are not modelled.
- **Workers' Comp:** Removed ~2.5% of records from provided data through cleaning steps, including removal of invalid observations, removal of NA rows, missing rows, or corrupted cells. Claim_count only ranges from 0 to 2, giving a compressed frequency distribution. Claim_length also has a finite range. The values of claim_amount were given from a range of 5-170, only one claim fell in this range, while the others were in the hundreds of thousands. The range was ultimately not taken into account during the modelling process.
- **Business Interruption:** Correlation between risks are not modelled, historical dataset may not capture future operational changes which could affect future claim patterns, ~2.5% of records removed from data cleaning process, which involved filtering out outliers, removal of NA rows, missing rows, or corrupted cells, there is an absence of insured value or sum insured, resulting in premiums being estimated on a per-station exposure basis rather than as a per unit of insured value.

8.3 Use of AI Tools

Employed AI tools to aid code creation - specifically for generating visualisations and plots via matplotlib.

9. Appendix

Appendix 1: Equipment Failure

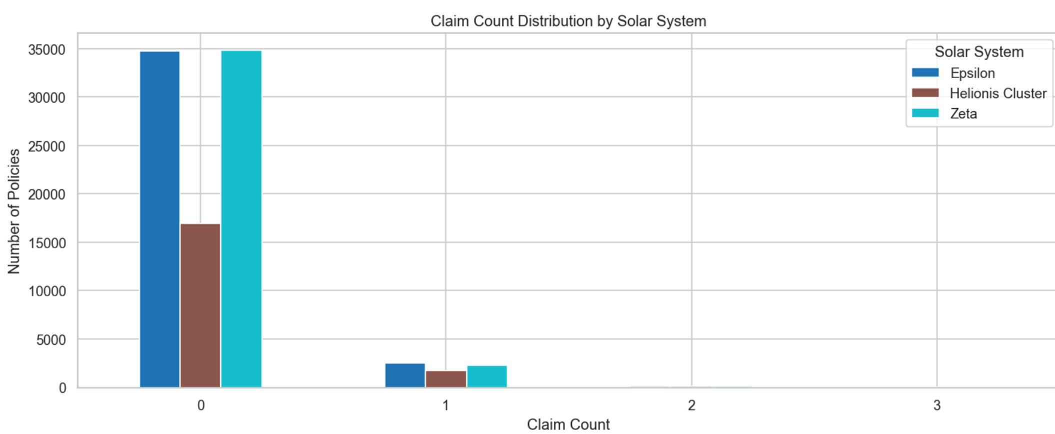
1. Data Cleaning and Preprocessing

- Removed missing values using df = df.dropna()
- Ensured numerical variables were correctly formatted
- Checked for invalid values:
    - Negative claim amounts removed
    - Zero exposure observations reviewed or excluded
    - Capclaim amounts at 790K, which is the upper bound
    - Remove entries outside of defined bounds in the data dictionary

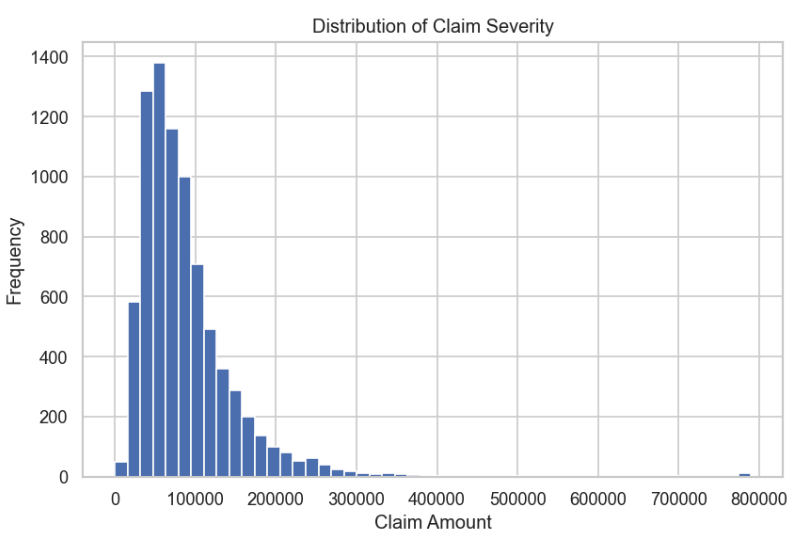
2. Exploratory Data Analysis (EDA)

2.1 Numeric Variable Distributions

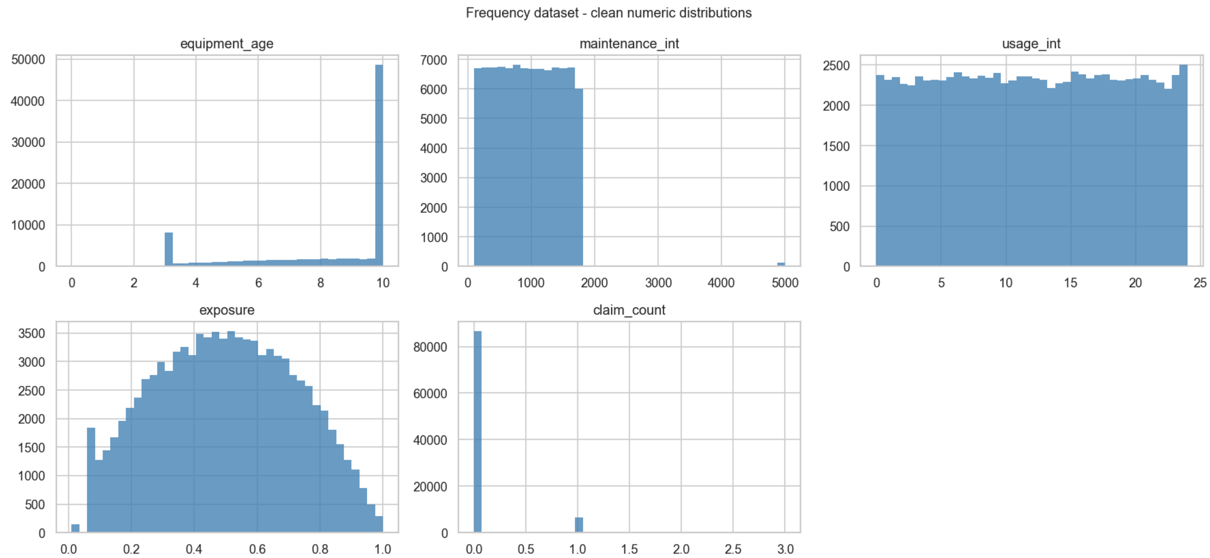
Claim Count distribution by Solar System



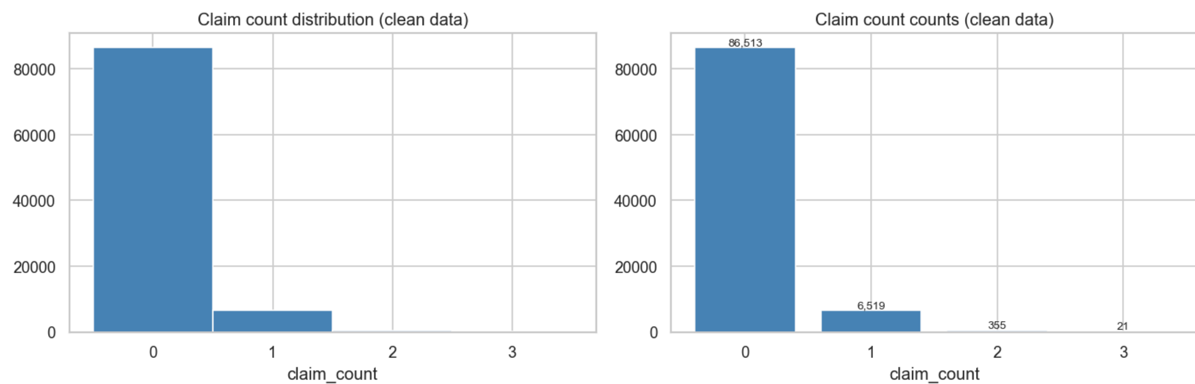
Histogram of the distribution of claim severity



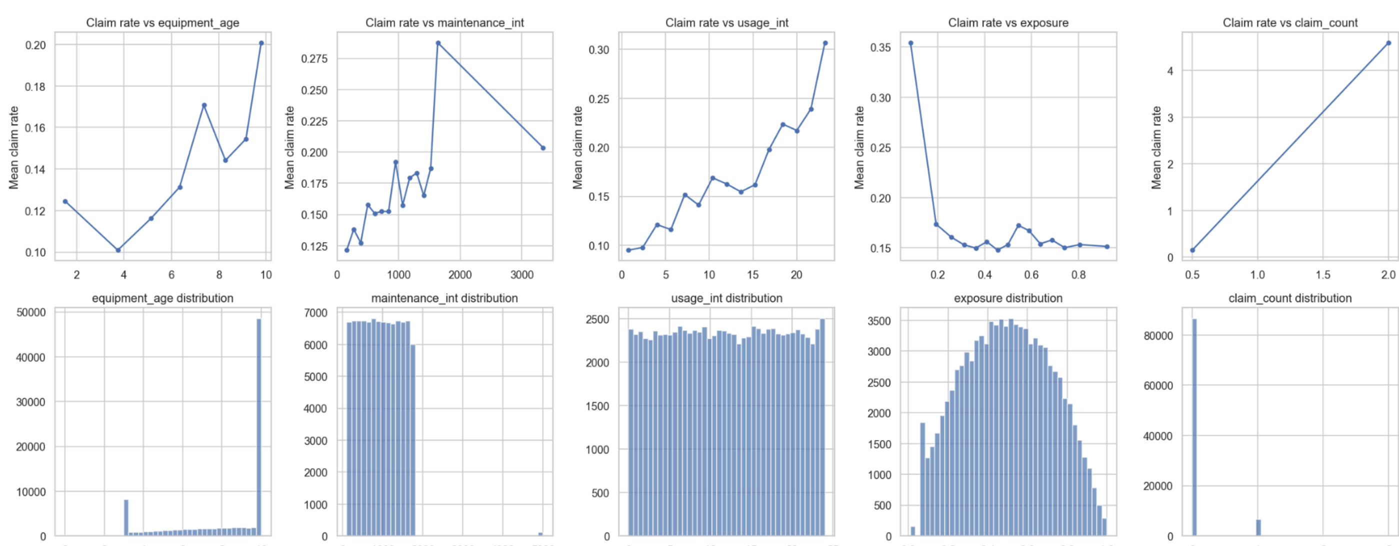
Numerical variables visualisation for frequency dataset



Claim Counts distribution



2-way variable comparison for numeric variables



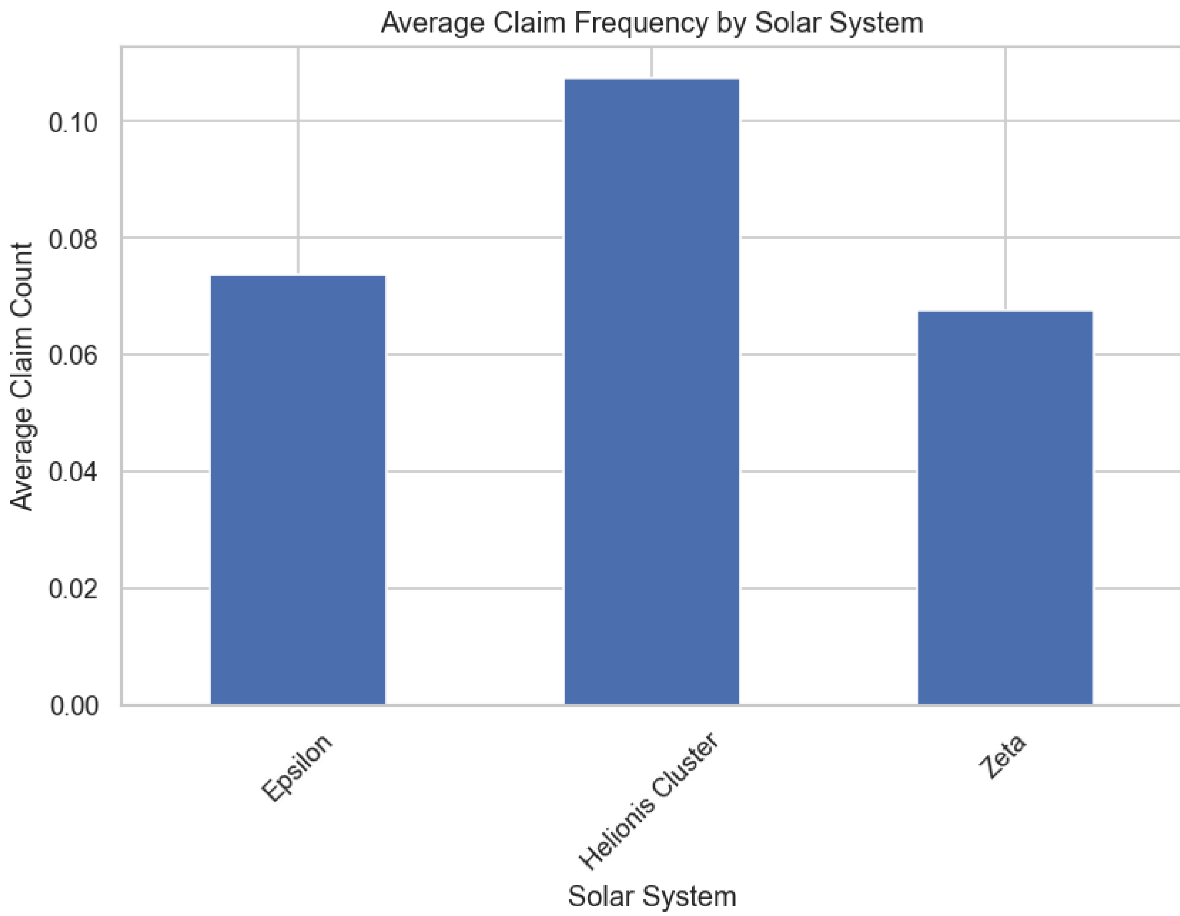
Correlation matrix for all numeric variables

Description:

- claim_count shows a **right-skewed discrete distribution**, consistent with count data
- Continuous variables show varying spread and potential skewness
- Severity exhibits **heavy right tail**, indicating large but infrequent losses

2.2 Claim Frequency by Solar System

Average claim frequency by solar system column graph

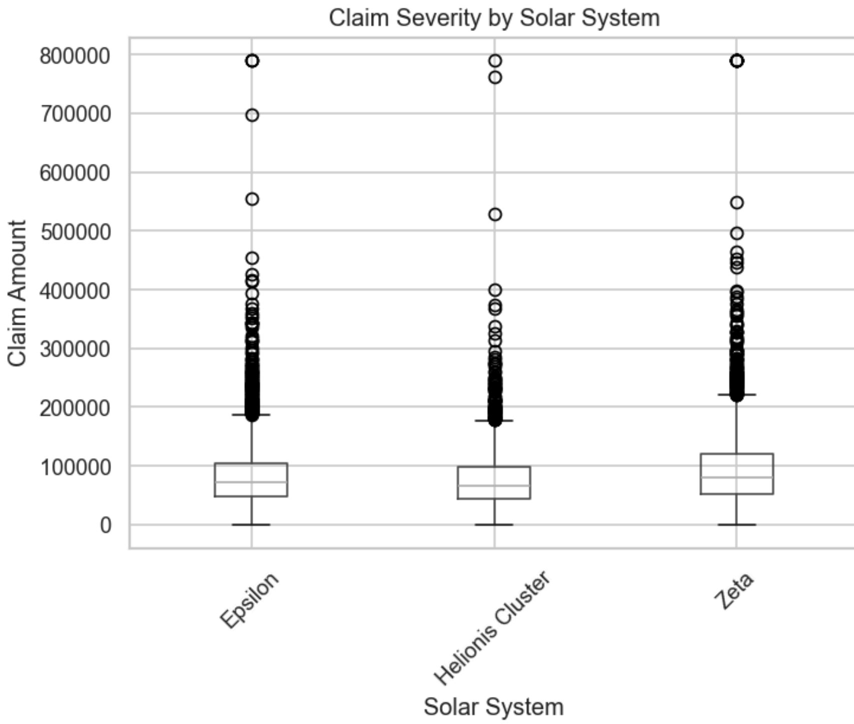


Description:

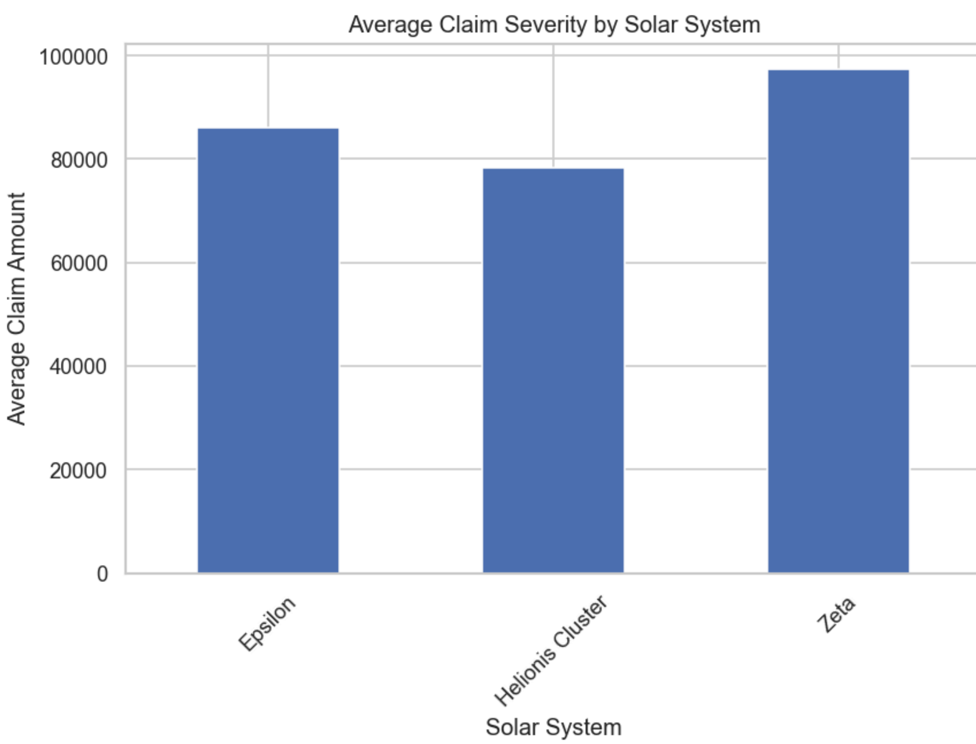
- Displays average claim count across solar systems
- Differences suggest **heterogeneity in risk exposure or operating conditions**
- Some systems exhibit elevated claim frequency, indicating higher operational risk

2.3 Claim Severity by Solar System

Box plot for claim severity by solar system



Average claim severity by solar system column graph

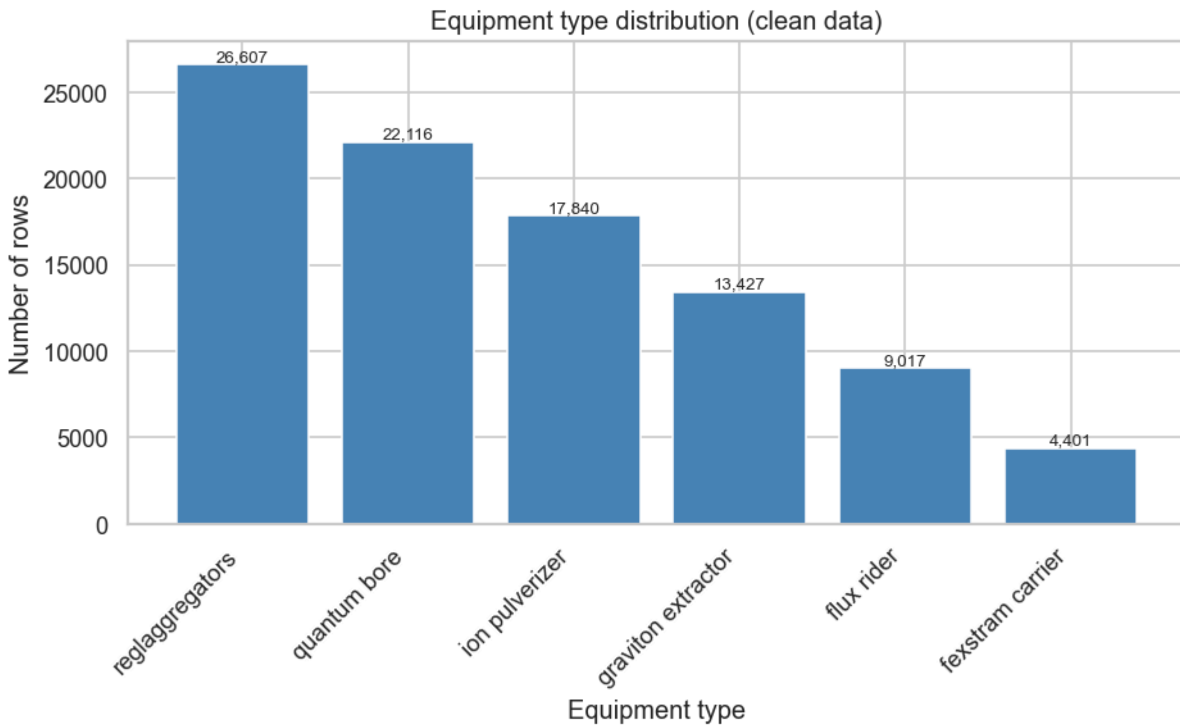


Description:

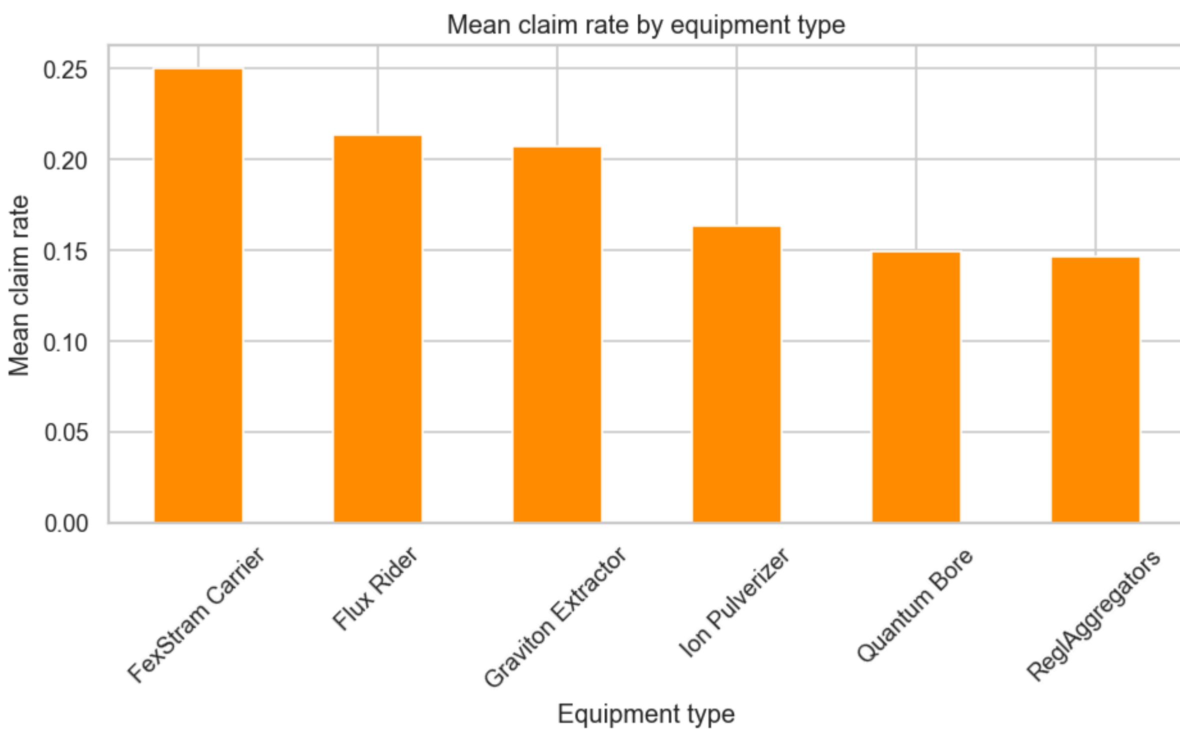
- Shows average claim size per solar system
- Significant variation suggests **location-specific loss behaviour**
- High severity systems may indicate **catastrophic or high-cost failures**

2.4 Categorical Variable Analysis

Column graph for the distribution of equipment type



Column graph for claim rate by equipment type



3. Model Specification

3.1 Frequency Model

A **Poisson Generalised Linear Model (GLM)** was used:

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \log(\text{exposure}_i)$$

Where:

- Y: claim count
- lambda: expected claim frequency
- Beta: covariates
- Exposure is included as an **offset term**

3.2 Severity Model

A **Gamma GLM with log link** was used:

$$Y_i \sim \text{Gamma}(\mu_i, \phi)$$

$$\log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Where:

- Y: claim amount
- mu: expected severity
- phi: dispersion parameter

4. Model Results

4.1 Frequency Model Results

Metric	Value
Number of Observations	93,408
Model Type	Poisson GLM
Link Function	Log
Log-Likelihood	-24,779
Deviance	35,530

Pearson Chi-Square	102,000
Pseudo R^2 (CS)	0.0146
Iterations	6

Variable	Mean	Std Dev
Equipment Age	8.1834	2.4432
Maintenance Interval	956.2496	513.8354
Usage Intensity	12.0088	6.9350

Frequency Model coefficients, p-value and interpretation

Variable	Coefficient	Exp(Coef)	P-value	Interpretation
Intercept	-1.6930	0.184	<0.001	Baseline claim frequency
Flux Rider	0.4084	1.504	<0.001	↑ frequency by ~50%
Graviton Extractor	0.1726	1.188	<0.001	↑ frequency by ~19%
Ion Pulverizer	0.1069	1.113	0.002	↑ frequency by ~11%
Quantum Bore	-0.0218	0.978	0.510	Not significant
Solar System: Epsilon	-0.3802	0.684	<0.001	↓ frequency by ~32%
Maintenance Interval (z)	0.1083	1.114	<0.001	↑ frequency with maintenance spacing

Model Deviance: 35530.0105
Residual DF: 93398
Dispersion (Deviance / DF): 0.3804
Pearson Dispersion: 1.0956

Interpretation:

- Positive coefficients → increase in claim frequency
- Negative coefficients → decrease in risk
- Solar system effects indicate geographical/systematic risk variation

The difference between deviance-based and Pearson-based dispersion measures suggests that while the model fits well overall, there may be minor deviations in variance structure. This is common in large datasets and does not materially impact model validity.

4.2 Severity Model Results

Metric	Value
Number of Observations	7,992
Model Type	Gamma GLM
Link Function	Log
Log-Likelihood	-95,038
Deviance	1,846.1
Pearson Chi-Square	2,000
Pseudo R^2 (CS)	0.2962
Scale Parameter	0.2510
Gamma Dispersion	0.2313
Iterations	9

Severity model coefficients, p-value and interpretation

Variable	Coefficient	Exp(Coef)	P-value	Interpretation
Intercept	10.7841	48,442	<0.001	Baseline claim size
Flux Rider	0.2048	1.227	<0.001	↑ severity by ~23%
Graviton Extractor	0.4214	1.524	<0.001	↑ severity by ~52%
Ion Pulverizer	0.5458	1.726	<0.001	↑ severity by ~73%
Quantum Bore	0.8935	2.444	<0.001	↑ severity by ~144%
Reglaggregators	0.5202	1.682	<0.001	↑ severity by ~68%
Solar System: Helionis Cluster	-0.0974	0.907	<0.001	↓ severity by ~9%
Solar System: Zeta	0.1148	1.122	<0.001	↑ severity by ~12%
Equipment Age (z)	0.0507	1.052	<0.001	↑ Severity with age
Usage Intensity (z)	0.1407	1.151	<0.001	↑ severity with usage

Metric	Value
Deviance	1,846.1
Residual DF	7,982
Deviance / DF	0.231
Pearson Dispersion	0.251

Interpretation:

- Coefficients interpreted multiplicatively due to the log link
- Larger coefficients → higher expected claim size

The severity model demonstrates substantially higher explanatory power than the frequency model, which is consistent with typical insurance modelling where claim sizes are more predictable than claim occurrences.

Since the dispersion is less than 1, this may indicate slight underdispersion or that the Gamma model fits the data particularly well. In practice, this is not problematic and supports the model choice.

5. Aggregate Loss Modelling

Aggregate loss simulated as:

$$L = \sum_{i=1}^{N} X_i$$

Where:

- $N \sim \text{Poisson}(\lambda)$
- $X_i \sim \text{Gamma}$

Simulation steps:

- Simulate frequency using Poisson distribution
- Simulate severity using Gamma distribution
- Sum losses to obtain total loss distribution

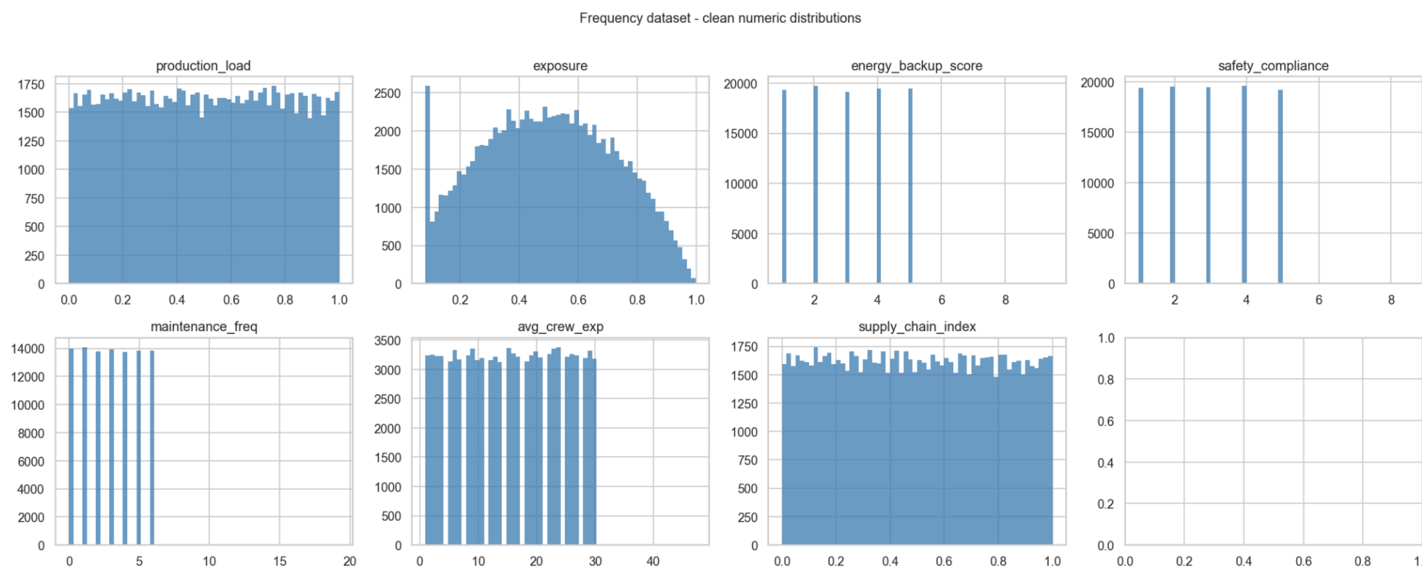
Business Interruption

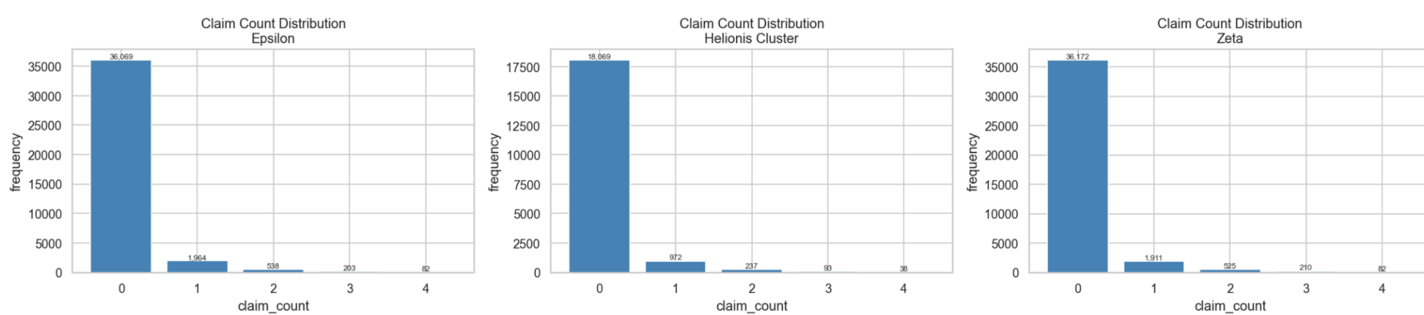
1. Data Cleaning and Preprocessing

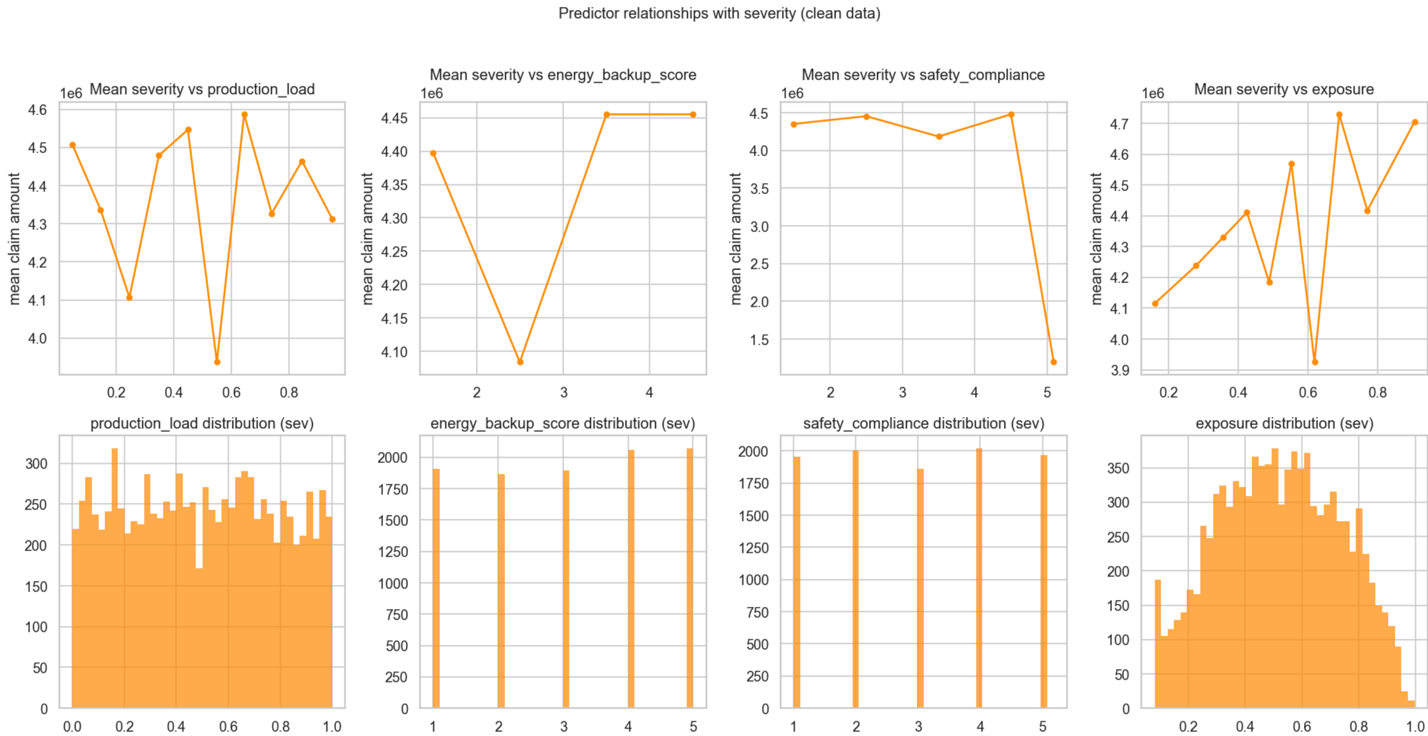
The business interruption dataset was preprocessed similar to the other datasets. First, both the frequency and severity datasets filtered for missing values present and then were removed. Records were also then audited using valid solar system labels, acceptable numeric ranges, and operational variables, with non-negative integer claim amounts for frequency, and strictly positive claim amounts for severity. Records that failed any validation were removed. Remaining rows with missing key modelling variables were then dropped. For modelling, categorical values were standardised by trimming whitespace and converting labels to lowercase, and ordinal predictors were encoded as ordered categories. Product segmentation for energy backup score was done and was grouped into weak, moderate, and strong tiers to form pricing classes for the solar systems.

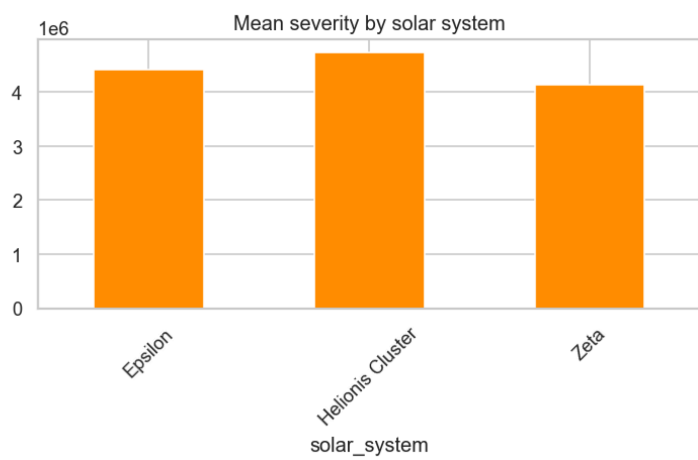
2. Exploratory Data Analysis (EDA)

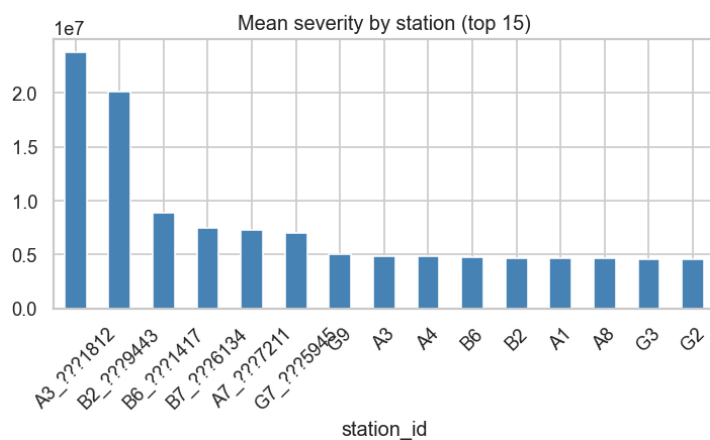
2.1 Numeric Variable Distributions



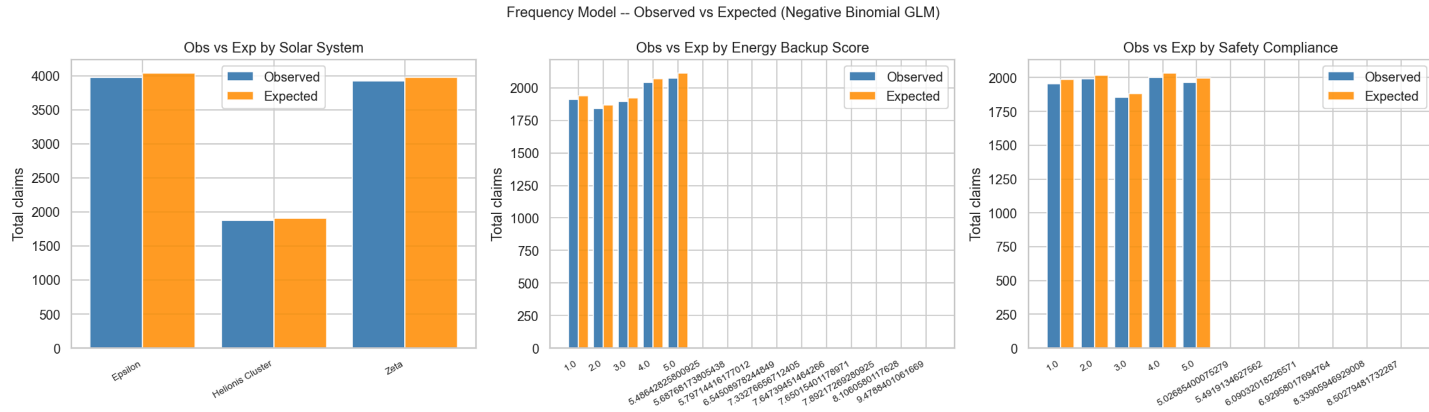




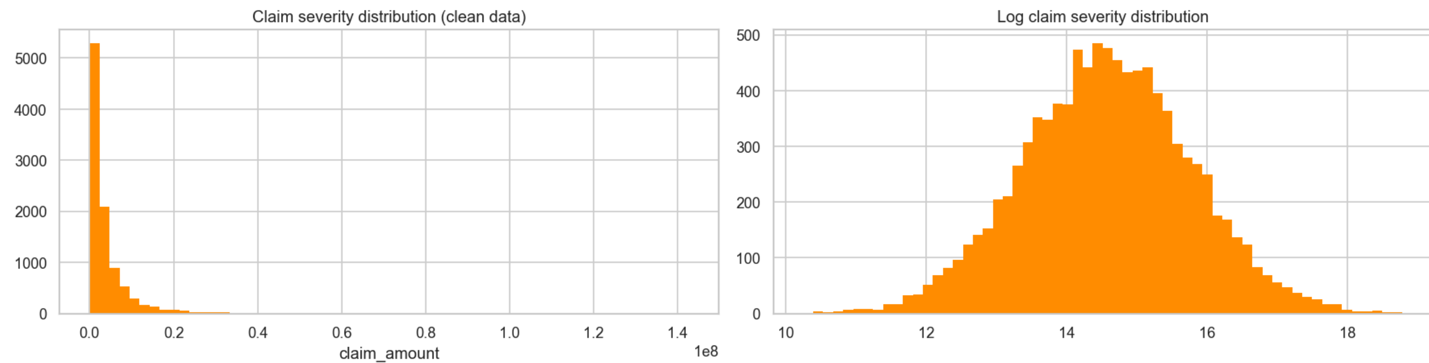




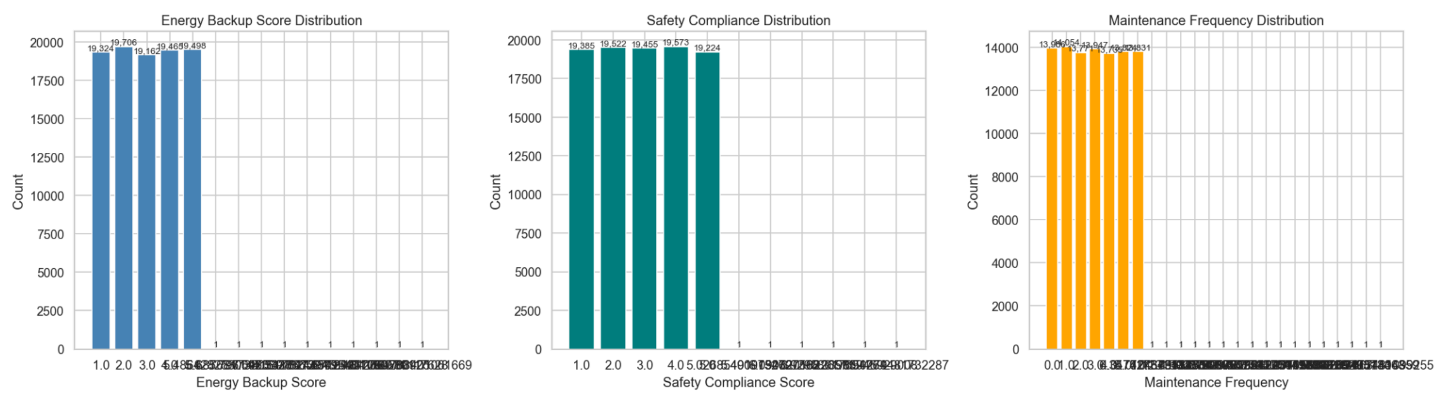
2.2 Claim Frequency by Solar System

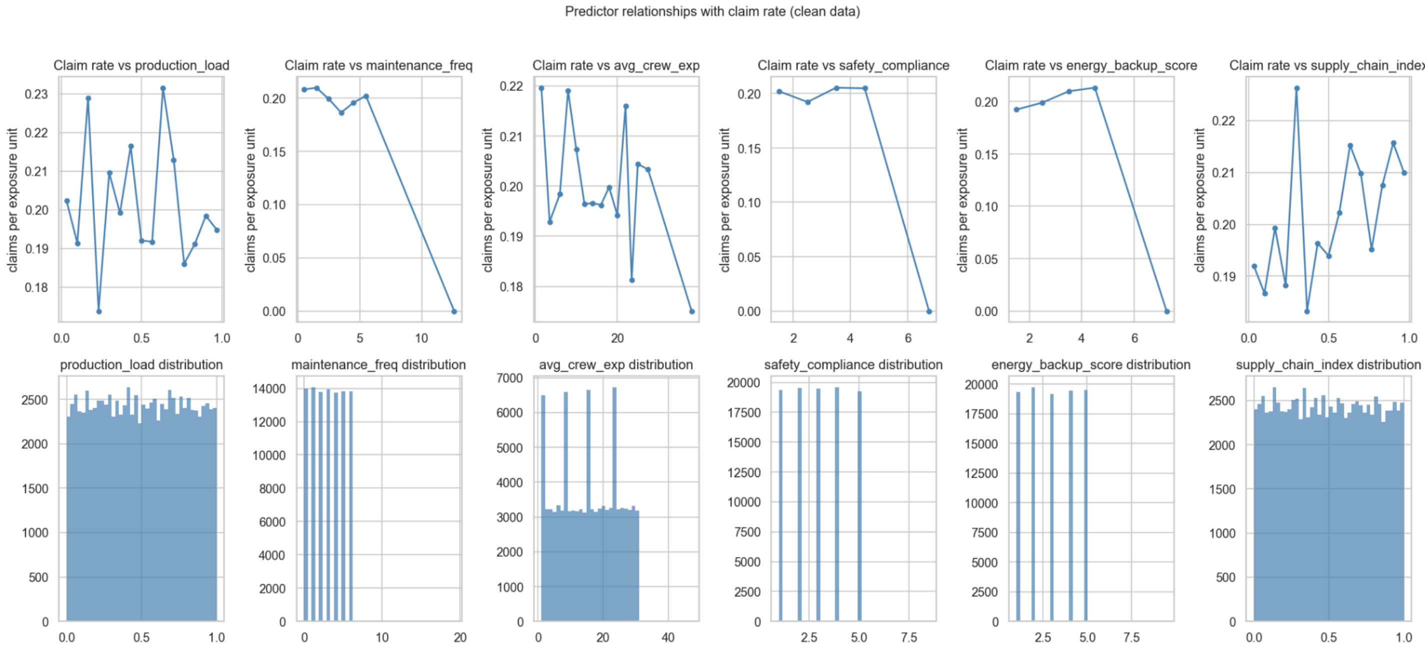


2.3 Claim Severity by Solar System

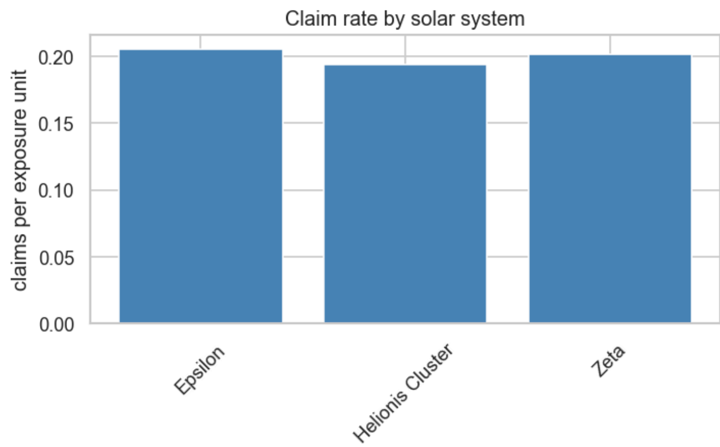


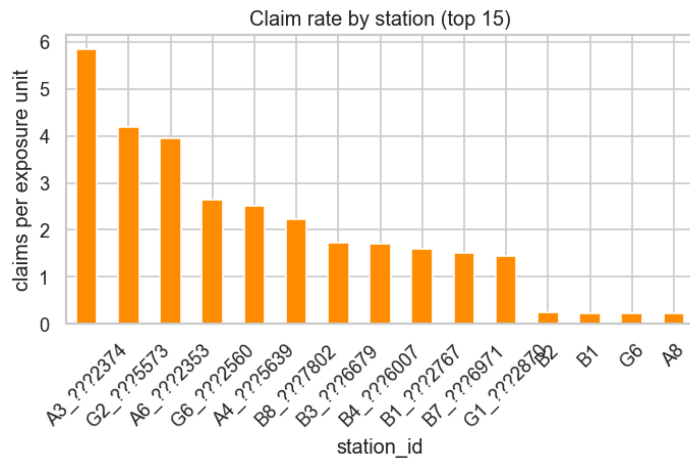
2.4 Categorical Variable Analysis

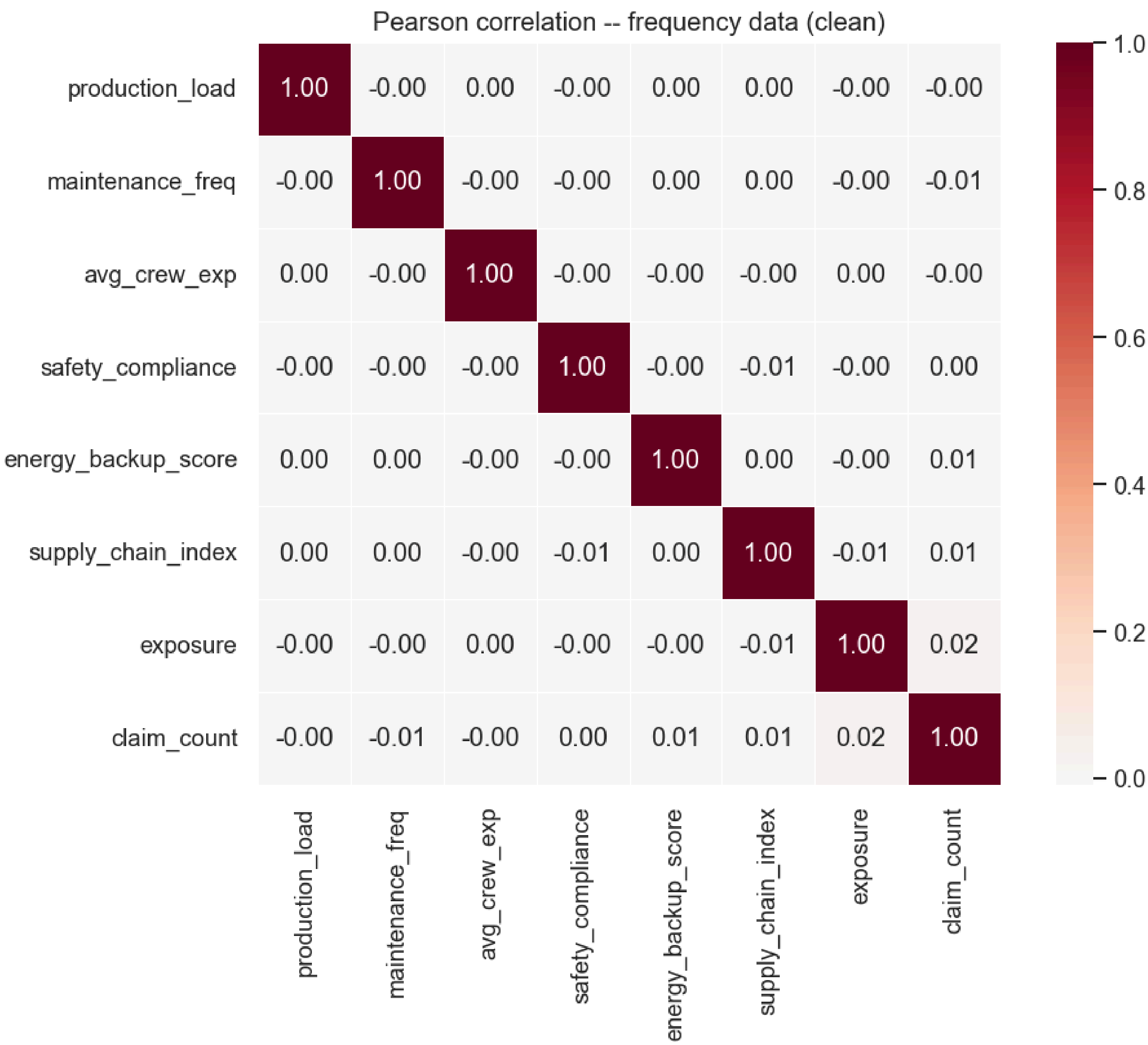




Predictor relationships with claim rate (clean data)
Claim rate vs production_load
Claim rate vs maintenance_freq
Claim rate vs avg_crew_exp
Claim rate vs safety_compliance
Claim rate vs energy_backup_score
Claim rate vs supply_chain_index
claims per exposure unit
production_load distribution
maintenance_freq distribution
avg_crew_exp distribution
safety_compliance distribution
energy_backup_score distribution
supply_chain_index distribution


Claim rate by solar system
claims per exposure unit
0.20
0.15
0.10
0.05
0.00
Epsilon
Helionis Cluster
Zeta


Claim rate by station (top 15)
claims per exposure unit
6
5
4
3
2
1
0
A3_???2374
G2_???5573
A6_???2353
G6_???2560
A4_???5639
B8_???7802
B3_???6679
B4_???6007
B1_???2767
B7_???6971
G1_???2870
B2
B1
G6
A8
station_id


Pearson correlation -- frequency data (clean)
production_load 1.00 -0.00 0.00 -0.00 0.00 0.00 -0.00 -0.00
maintenance_freq -0.00 1.00 -0.00 -0.00 0.00 0.00 -0.00 -0.01
avg_crew_exp 0.00 -0.00 1.00 -0.00 -0.00 -0.00 0.00 -0.00
safety_compliance -0.00 -0.00 -0.00 1.00 -0.00 -0.01 -0.00 0.00
energy_backup_score 0.00 0.00 -0.00 -0.00 1.00 0.00 -0.00 0.01
supply_chain_index 0.00 0.00 -0.00 -0.01 0.00 1.00 -0.01 0.01
exposure -0.00 -0.00 0.00 -0.00 -0.00 -0.01 1.00 0.02
claim_count -0.00 -0.01 -0.00 0.00 0.01 0.01 0.02 1.00
production_load
maintenance_freq
avg_crew_exp
safety_compliance
energy_backup_score
supply_chain_index
exposure
claim_count
1.0
0.8
0.6
0.4
0.2
0.0

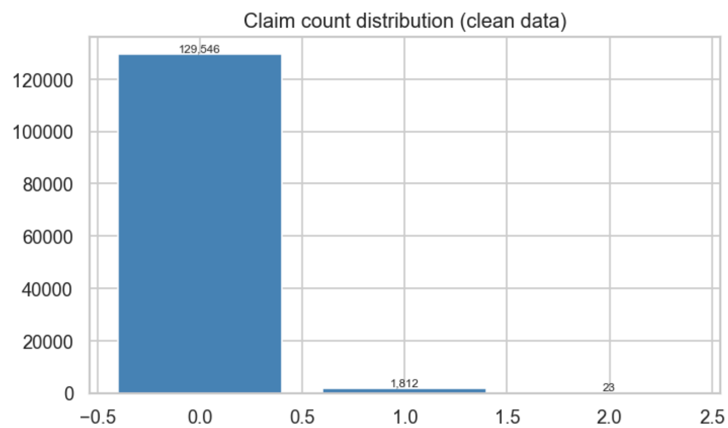
Workers Compensation

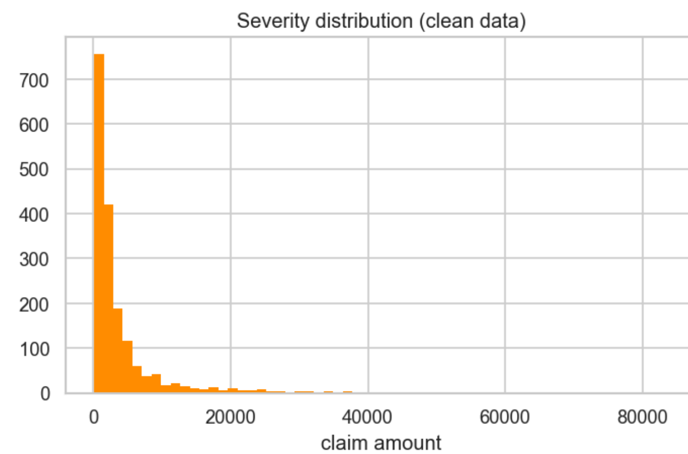
1. Data Cleaning and Preprocessing

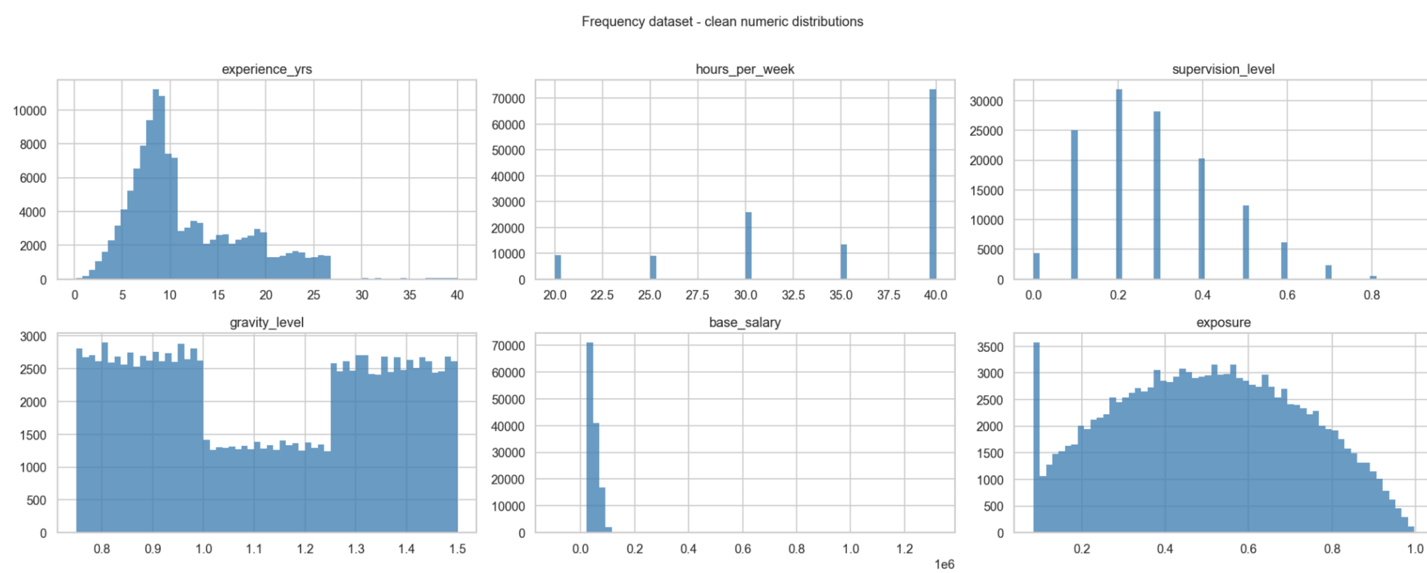
- Removed missing values
- Ensured numerical variables were correctly formatted
- Checked for invalid values:
  - Negative claim amounts removed
  - Zero exposure observations reviewed or excluded
  - Cap claim amounts at 170, which is the upper bound
  - Remove entries outside of defined bounds in the data dictionary

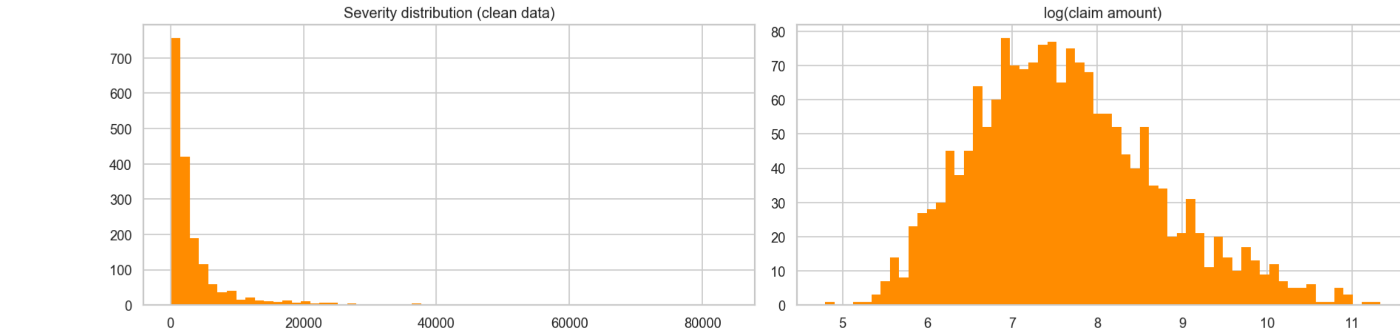
2. Exploratory Data Analysis (EDA)

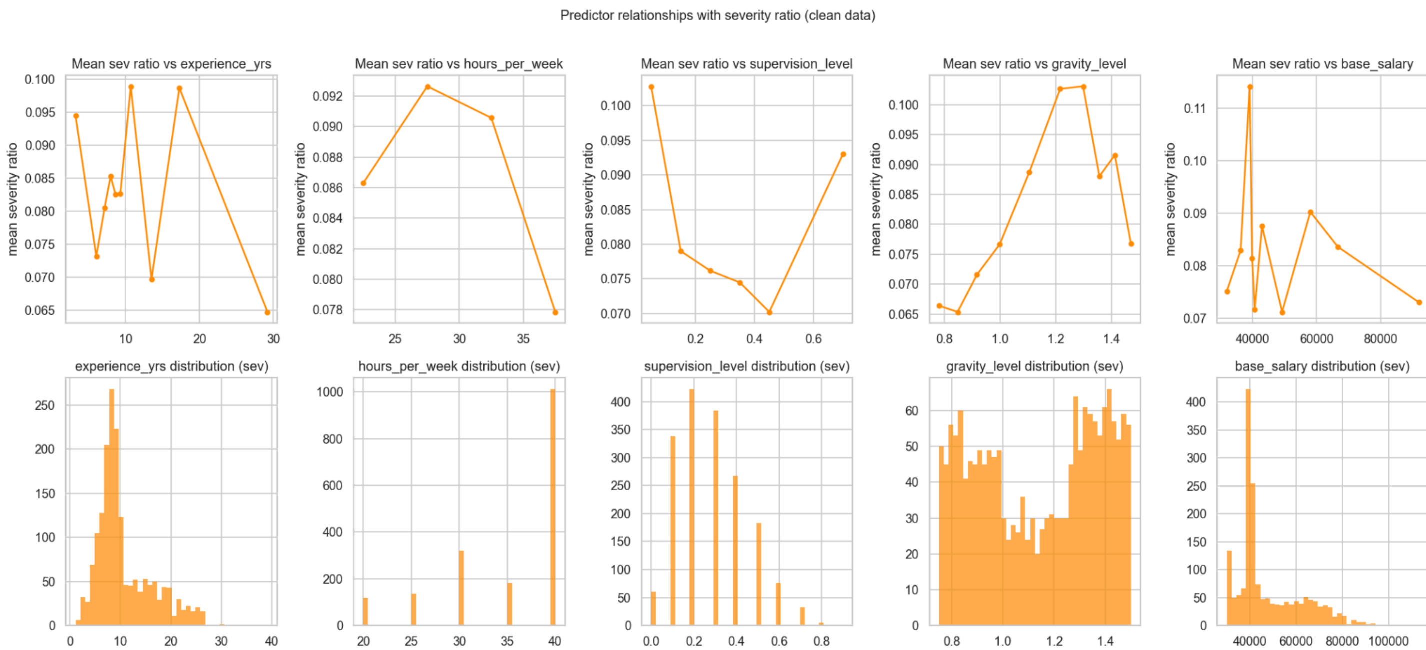
2.1 Numeric Variable Distributions

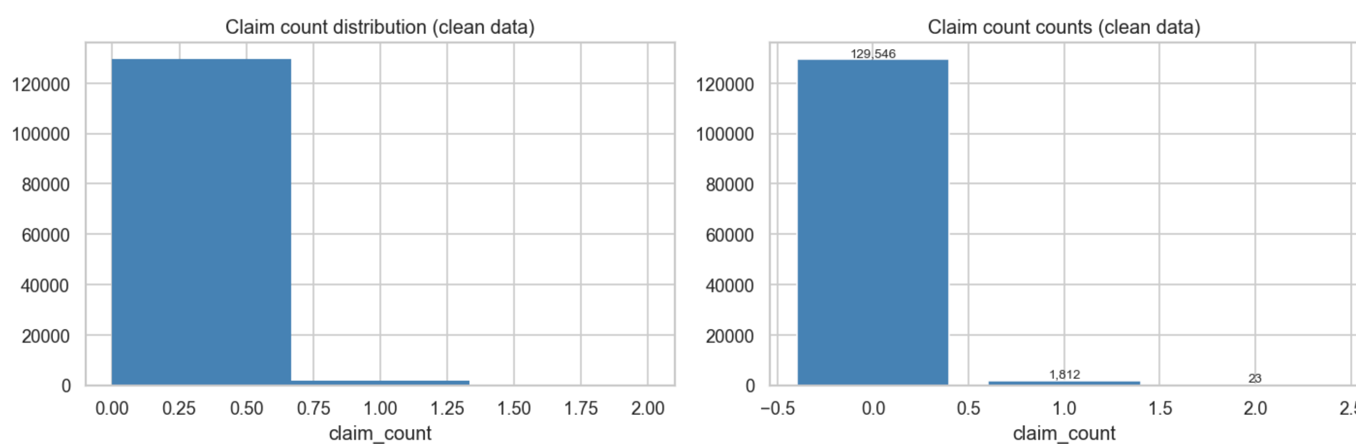


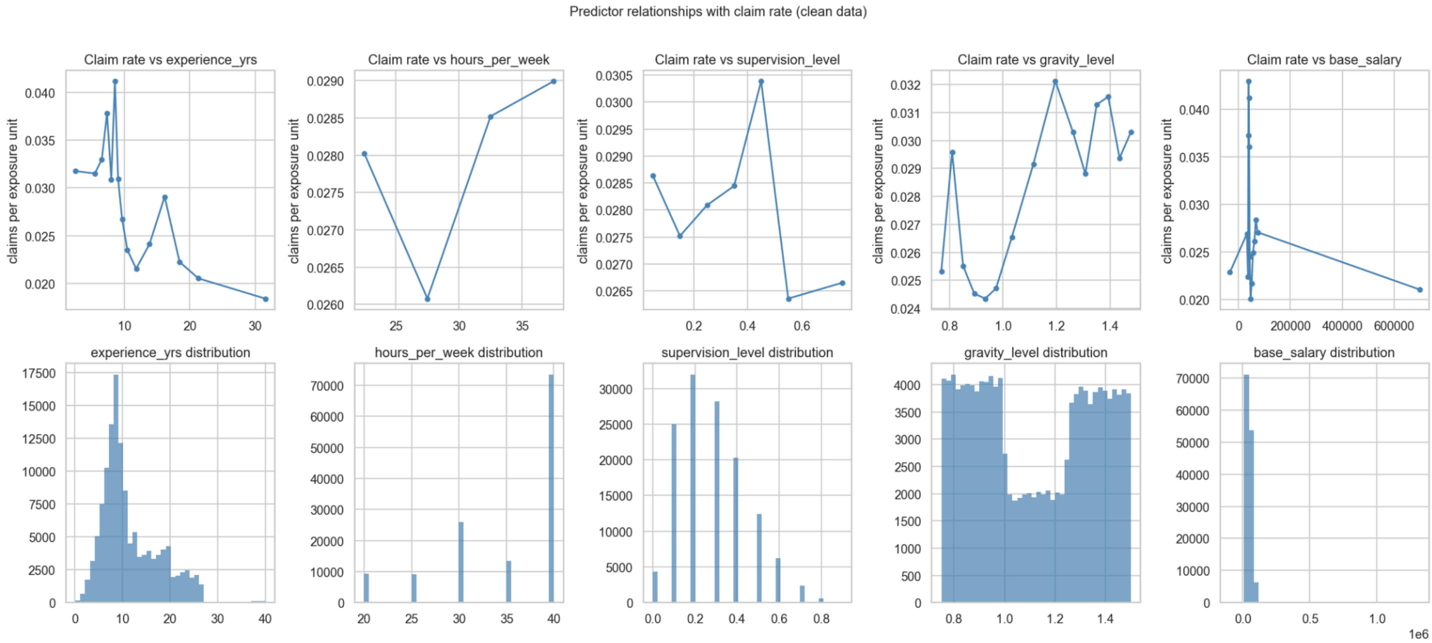




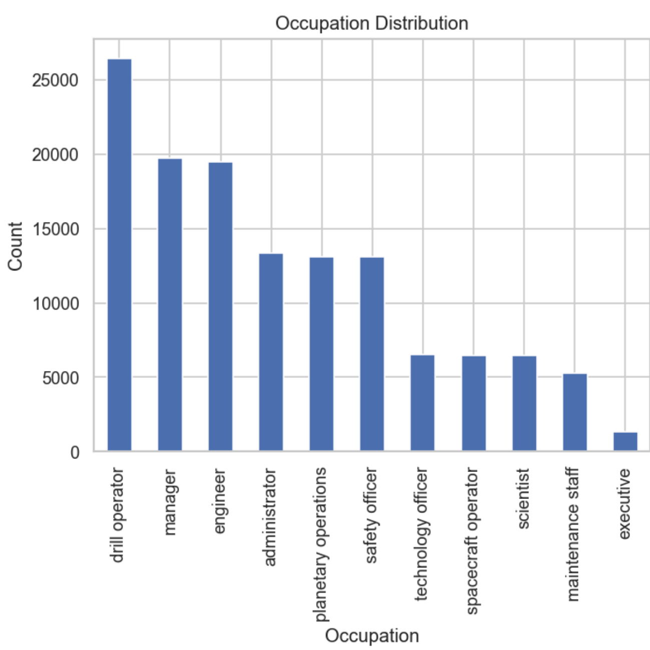

Severity distribution (clean data)
log(claim amount)

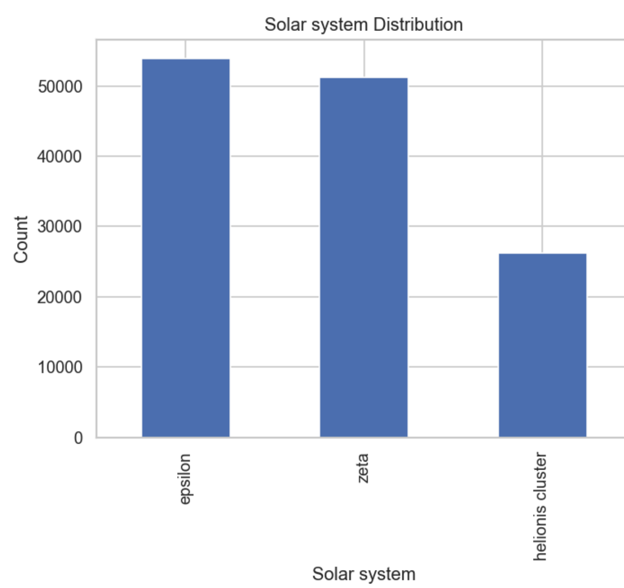

Predictor relationships with severity ratio (clean data)
Mean sev ratio vs experience_yrs
Mean sev ratio vs hours_per_week
Mean sev ratio vs supervision_level
Mean sev ratio vs gravity_level
Mean sev ratio vs base_salary
mean severity ratio
experience_yrs distribution (sev)
hours_per_week distribution (sev)
supervision_level distribution (sev)
gravity_level distribution (sev)
base_salary distribution (sev)

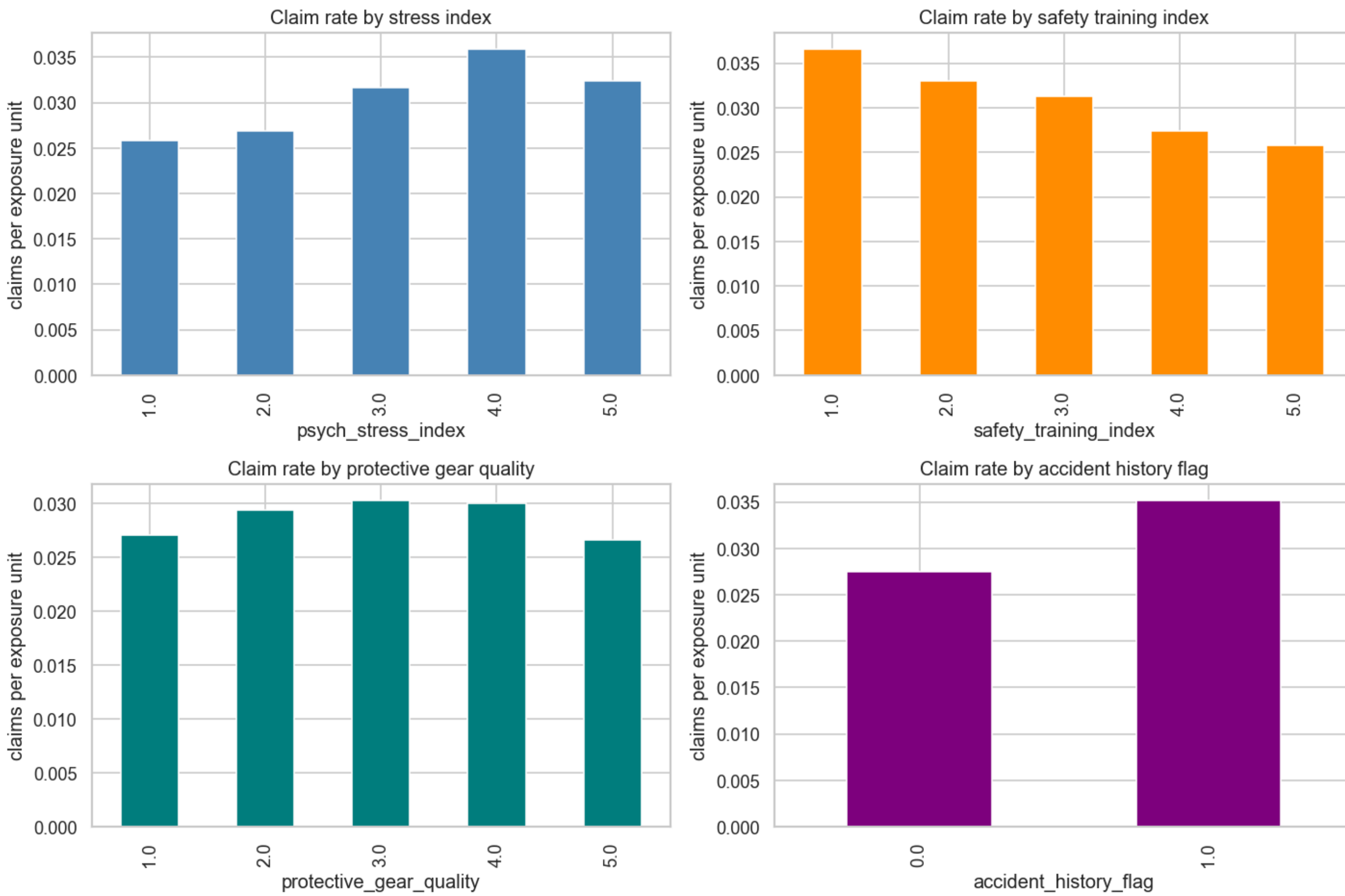

Claim count distribution (clean data)
claim_count
Claim count counts (clean data)
129,546
1,812
23
claim_count



2.2 Categorical Variable Analysis







3. Model Specification

3.1 Frequency Model

A **Poisson Generalised Linear Model (GLM)** was used:

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \log(\text{exposure}_i)$$

Where:

- Y: claim count
- lambda: expected claim frequency
- Beta: covariates
- Exposure is included as an **offset term**

3.2 Severity Model

A **Log-Normal Ordinary Least Squares Regression** was used:

$$\log(Y_i) \sim \mathcal{N}(\mu_i, \sigma^2)$$

$$\mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Where:

- Y: claim amount
- mu: expected log severity
- sigma-squared: variance of log severity
- X: covariates
- Beta: coefficients

4. Model Results

4.1 Frequency Model Results

Metric	Value
Number of Observations	131,142
Model Type	Poisson GLM
Link Function	Log
Log-Likelihood	-9,392.4
Deviance	15,103
Pearson Chi-Square	130,000
Pseudo R^2 (CS)	0.00249
Iterations	7

Frequency Model coefficients, p-value and interpretation

Variable	Coefficient	Exp(Coef)	P-value	Interpretation
Intercept	-4.334	0.013	<0.001	Baseline claim frequency is low
Drill Operator	1.136	3.11	<0.001	~3x higher frequency

Engineer	0.656	1.93	<0.001	~93% higher frequency
Maintenance Staff	0.723	2.06	<0.001	~ 2x higher frequency
Technology Officer	0.702	2.02	<0.001	~2x higher frequency
Accident History	0.244	1.28	<0.001	Increases frequency by ~ 28%
Stress Index	0.087	1.09	<0.001	Increases frequency per level
Safety Training	-0.089	0.91	<0.001	Decreases frequency by ~ 9% per level

Model Deviance: 15,103
Residual DF: 13119
Dispersion (Deviance / DF): 0.115
Pearson Dispersion: 0.99

Interpretation:

- Positive coefficients → increase in claim frequency
- Negative coefficients → decrease in risk

The difference between deviance-based and Pearson-based dispersion measures suggests that while the model fits well overall, there may be minor deviations in variance structure. This is common in large datasets and does not materially impact model validity.

4.2 Severity Model Results

Metric	Value
Number of Observations	1,770
Model Type	Log-Normal
F-statistic	36.67
R-Squared	0.226
Adjusted R-Squared	0.22
AIC	4.866
BIC	4,948
Mu	7.6198
Sigma-squared	1.1633

Severity model coefficients, p-value and interpretation

Variable	Coefficient	Exp(Coef)	P-value	Interpretation
Intercept	7.423	1,676.5	<0.001	Baseline claim severity
Engineer	0.573	1.774	<0.001	Increase severity by ~77%
Executive	0.718	2.051	0.040	Increase severity by ~105%
Manager	0.447	1.564	<0.001	Increase severity by ~56%
Scientist	0.624	1.867	<0.001	Increase severity by ~87%
Technology Officer	0.300	1.350	0.027	Increase severity by ~35%
Psych Stress Index	0.289	1.335	<0.001	Increases severity by ~ 33% per unit
Gravity Level	0.372	1.451	<0.001	Higher gravity increases severity by ~45%
Safety Training Index	-0.142	0.868	<0.001	Decreases severity by ~13%
Protective Gear Quality	-0.140	0.870	<0.001	Decreases severity by ~13%

Interpretation:

- Larger coefficients leads to higher expected claim size
- Severity model is mainly driven by factors relating to the workplace environment of a particular occupation such as stress or safety.
- Executives had claims with larger claim amounts
- Severity was right-skewed meaning the mean claim amount is greater than median which makes the data a good match for a log-normal distribution.